# BioCoLoop: Collaborative Agentic Research for Biological Model Improvement

**Anonymous authors**
Paper under double-blind review

## Abstract

Biological model development requires both fitting parameters and deciding which model configurations to investigate. When measurements are held by different laboratories, these decisions must draw on distributed training and evaluation. We introduce BioCoLoop, a collaborative research framework that connects laboratory-local fitting to a shared record of candidate designs, development trajectories and selection decisions. Task adapters implement this interface for drug–target interaction prediction, observed-response proteomic efficacy prediction and cell perturbation identification. We evaluate three distinct uses of collaboration: increasing laboratory participation, conditioning proposals on prior development outcomes, and allocating training from early learning trajectories. Source-defined laboratory experiments measure transfer across biological studies, while same-access search controls separate proposal quality from access to additional data. The combined framework improves four of five primary endpoints relative to the single-laboratory fixed model; the effect of proposal history is task-dependent. A separate allocation study improves double-gene perturbation identification by 10.84 percentage points over uniform allocation at matched training and evaluation counts, although simpler promotion also recovers much of this gain. These results establish a common experimental framework for collaborative biological model development and identify training allocation and development-to-test generalization as consequential research decisions.

## 1. Introduction

Artificial intelligence supports biological research through predictive modeling and the generation and evaluation of hypotheses (Cui et al., 2024; Theodoris et al., 2023; Roohani et al., 2024). Self-improving agents automate parts of model development (Jiang et al., 2025b; Lu et al., 2024; Zhou et al., 2026) through an iterative process of proposing, testing and revising (Madaan et al., 2023; Shinn et al., 2023; Huang et al., 2023). An agent proposes changes to a model design or its training hyperparameters, trains and evaluates the candidate, and uses the results to inform the next proposal.

In biology, useful data are often distributed across laboratories studying different cell types, perturbations or experimental assays. Data-sharing constraints can prevent these measurements from being pooled (Sheller et al., 2020; Bakhtiari et al., 2025; Dayan et al., 2021; Kaissis et al., 2020; Rieke et al., 2020). Collaborative learning enables laboratories to train a shared model through locally computed updates (McMahan et al., 2017; Wang et al., 2024; Li et al., 2020; Karimireddy et al., 2020; Kairouz et al., 2021). Automated federated design already combines configuration or system changes with distributed fitting (Khodak et al., 2021; Li et al., 2026). The biological question is how to connect these decisions to measurements from different assays and experimental contexts, and how to distinguish the value of additional data from the value of the research procedure.

Local development evidence provides a complementary research signal: it records how candidate configurations perform, how their training trajectories evolve and which trials succeed or fail. Sharing these summaries allows laboratories to inform subsequent experiments while retaining their measurements locally. *Can an iterative research process use laboratory-local training and evaluation to guide model development without pooling the underlying data?*

BioCoLoop extends collaboration from parameter fitting to model development by sharing both model updates and laboratory-local development evidence. Figure 1 places this framework along-

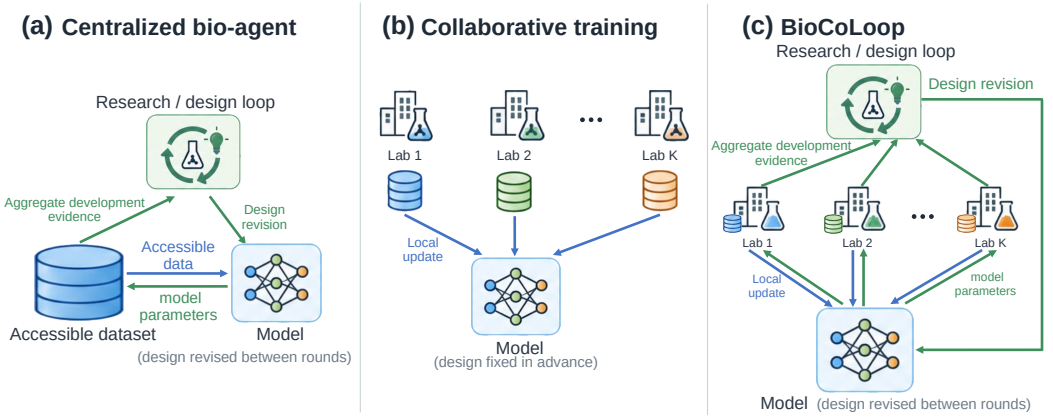



Figure 1: Research settings for biological AI. A centralized bio-agent revises a model using centrally accessible data. Collaborative training allows $K$ laboratories to optimize the same model design and training hyperparameters by sharing model updates rather than raw data. BioCoLoop additionally uses results from local evaluation to guide subsequent changes to the model design and training hyperparameters.

side centralized self-improving agents and conventional collaborative training. An inner loop fits and evaluates candidates across laboratories. An outer loop uses the resulting evidence to revise proposals. Collaborative participation supplies training data and evaluation panels; development-history feedback uses their measured outcomes to guide search. We also evaluate a separate policy that uses early development trajectories to allocate further training.

We evaluate BioCoLoop on three biological task families with two proposal models, conventional search controls and task-adapted research-agent baselines. The evaluation separates an access scenario—ten participating laboratories versus one—from same-access tests of search and allocation. Our contributions are:

- A common biological research interface connects local fitting, development evaluation and executable configuration proposals across interaction, proteomic and perturbation tasks.
- A shared experiment record supports two separately evaluated decisions: choosing the next configuration and allocating further training from early scores and improvement trajectories.
- Controlled experiments distinguish laboratory participation, cross-study transfer and same-access search, and identify both a matched-budget allocation gain and cases where development optimization fails to improve held-out prediction.

## 2. Related work

**Collaborative learning for biological data.** Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman synthesizes collaborative-learning systems through planning, coding and simulation-based refinement (Li et al., 2026).

**Self-improving agents in biological research.** Self-improving agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist and AI-Scientist-v2 connect ideas to executed experiments, replanning, tree search and writing (Lu et al., 2024; Yamada et al., 2025). DrugEvolve combines research, implementation, evaluation and algorithm memory for drug-

development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). BioCoLoop studies how multiple data-owning laboratories can supply development feedback through a shared research interface. TAPB, a ProteinTalks-derived efficacy head and scDEBART-based cell representations provide its task-specific prediction components (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). The proposal schema, fitting service and selection mechanism are common across tasks. Strong simple references on some perturbation benchmarks motivate evaluating the research procedure with fixed task predictors (Ahlmann-Eltze et al., 2025). Configuration search also connects to random search (Bergstra & Bengio, 2012), Bayesian optimization (Snoek et al., 2012) and language-model proposal distributions (Yang et al., 2024). We use feedback-free direct search to isolate the effect of development history under a shared proposal and fitting interface.

**Collaborative scientific agents.** The Virtual Biotech coordinates specialist agents around therapeutic-discovery evidence (Zhang et al., 2026). AutoScientists organizes experimentation through shared state and records successful and failed directions (Gao et al., 2026). Helmsman directly addresses agent-generated federated systems, including simulation-based diagnosis and revision (Li et al., 2026). BioCoLoop's focus is a common biological predictor-development interface: laboratories supply local fits and development measurements, and controlled experiments separate participation, proposal history and training allocation. Its finite configuration library supports these comparisons; it differs from open-ended federated-system code synthesis.

## 3. Method: collaborative biological model improvement

### 3.1 Problem setting and notation

Consider $K$ laboratories working on the same biological prediction task. Laboratory $i$ holds training data $D_i^{\text{train}}$, with $n_i = |D_i^{\text{train}}|$. Evaluation uses laboratory-local development panels $\{D_i^{\text{dev}}\}_{i\in\mathcal{P}}$, where $\mathcal{P}$ indexes the panels specified by the evaluation protocol. A task specification $\mathcal{T}$ defines the prediction inputs, outputs, training loss and evaluation metric. We write the task predictor as $f_{a,w}$, where $a \in \mathcal{A}$ specifies its design and $w$ its trainable parameters. Given an initial design, a design space $\mathcal{A}$ and a research budget, BioCoLoop seeks a design and fitted predictor that perform well on the designated evaluation population. The three task families instantiate $f_{a,w}$ as task-specific predictors within this shared research procedure.

In the main experiments, the *registered executable design space* $\mathcal{A}$ contains complete candidate configurations with four editable choices: learning rate, weight decay, coordinator momentum and an optional residual prediction module. Each `design_id` selects one configuration. The architecture choice enables or disables a width-32 residual module with a zero-initialized output layer; the task adapter supplies the backbone and prediction head. Appendix E.6 defines a separate, larger-data mechanism study with its own design space and training schedule.

A *shared coordinator* is the trusted execution service used by all participating laboratories. It resolves each design identifier, initializes and broadcasts the predictor, coordinates local fitting, aggregates model states and bounded development measurements, and applies the checkpoint and design-selection rules. Laboratories receive the same candidate state and schedule while retaining raw training and development measurements locally. The coordinator supplies the aggregate evidence to research model $S$ (Figure 2), a language model whose weights $\theta$ remain fixed throughout a run. We use this fitting interface to evaluate history-guided proposal revision (Section 3.3) and trajectory-guided training allocation (Section 3.4) separately.

### 3.2 Collaborative fitting and shared evaluation

For each candidate $a$, the coordinator resets the predictor to its prescribed initialization $w_0$ and distributes it to all training laboratories. Let $\text{LocalEpoch}_a$ denote one full local AdamW epoch with the candidate's learning rate and weight decay, batch size 64 and gradient-norm clipping at one. The optimizer state resets at each round. With sample weights $\pi_i = n_i / \sum_{j=1}^{K} n_j$, coordinator momentum

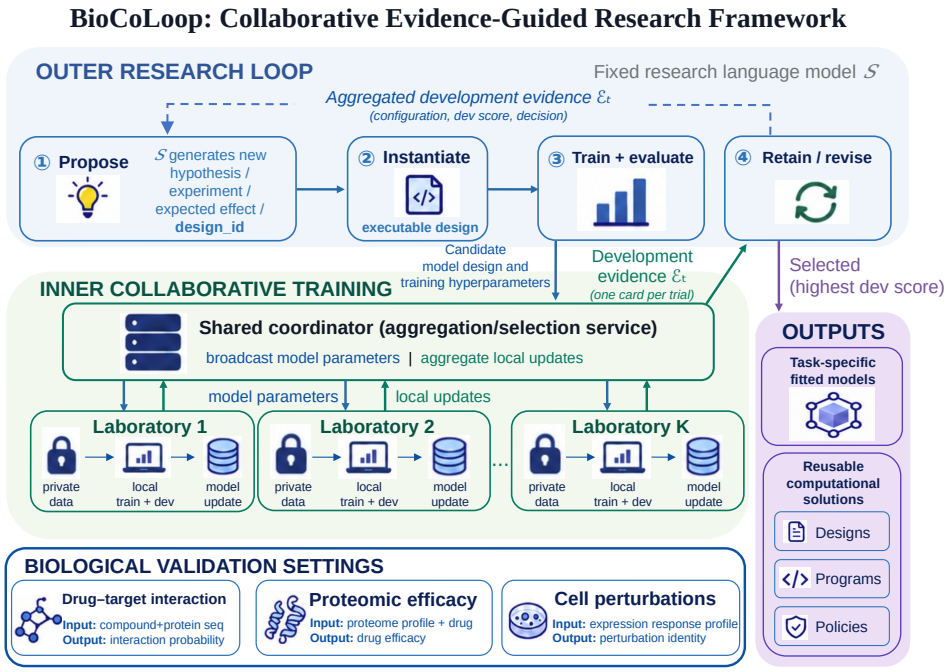



Figure 2: BioCoLoop's common research interface. Research model $S$ proposes an executable configuration; the coordinator fits it using updates from $K$ laboratories and summarizes development evaluation in an evidence card. The ordered history $\mathcal{E}_t$ conditions the next proposal. Task adapters supply biological inputs, prediction heads and losses. Sections 3.2–3.3 define fitting, evidence and selection; Section 3.4 evaluates training allocation separately.

coefficient $\beta_a$ and initial velocity $v_0 = 0$, round $r$ updates the trainable parameters by

$$u_{i,r} = \mathrm{LocalEpoch}_a(w_{r-1}; D_i^{\mathrm{train}}), \quad \bar{w}_r = \sum_{i=1}^{K} \pi_i u_{i,r}, \\ v_r = \beta_a v_{r-1} + \bar{w}_r - w_{r-1}, \qquad w_r = w_{r-1} + v_r. \tag{1}$$

The coordinator carries momentum across aggregation rounds through $v_r$. With $\beta_a = 0$, aggregation reduces to sample-weighted parameter averaging (McMahan et al., 2017). In each round, the coordinator sends the current trainable state and fixed candidate settings to every laboratory. Laboratory $i$ returns its locally fitted state $u_{i,r}$ and training-example count $n_i$. Frozen feature extractors remain unchanged. Task adapters supply the biological inputs, trainable prediction components and local loss; raw examples, expression profiles, labels and per-example identifiers remain local.

Every five rounds and at the final round, laboratories evaluate the aggregated predictor $f_{a,w_r}$ on the development panels indexed by $\mathcal{P}$. Each returns a scalar primary score $s_i(a, r)$, development loss $\ell_i^{\mathrm{dev}}(a, r)$ and the bounded trajectory diagnostics required by the protocol. The main comparison uses participating laboratories' panels; fixed-pool repartitioning retains all ten original panels; source-transfer studies use the original target panels. The coordinator computes

$$\bar{s}(a, r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} s_i(a, r), \qquad \bar{\ell}(a, r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \ell_i^{\mathrm{dev}}(a, r). \tag{2}$$

The selected round $\widehat{r}_a$ maximizes $\bar{s}(a, r)$, with lower $\bar{\ell}(a, r)$ breaking ties and the earlier round retained on a complete tie. Its parameters are $\widehat{w}_a = w_{\widehat{r}_a}$. The pair $q(a) = (\bar{s}(a, \widehat{r}_a), -\bar{\ell}(a, \widehat{r}_a))$ ranks the design. We call $\bar{s}(a, \widehat{r}_a)$ the *dev score*, labelled "highest dev score" in Figure 2. Training weights examples, while selection weights the designated development panels equally. The evaluation schedule assigns each candidate 100 training rounds in the main proposal-search experiments. Each trial produces a *development-evidence card*. After $t$ trials, the sequence $\mathcal{E}_t = (e_1, \ldots, e_t)$

forms the accumulated *development evidence*. For $1 \le j \le t$, card $e_j$ records the executed configuration $a_j$, hypothesis $h_j$, best development primary score and loss $(s_j, \ell_j)$, selected round $\widehat{r}_j$ and acceptance decision $\mathrm{dec}_j$. Endpoint diagnostics $\Delta_j$ contain the first and last development-evaluation rounds, their sample-weighted training losses, equal-laboratory development scores and losses, and the changes between those measurements, plus the optional worst-laboratory score and evaluating-laboratory count.

For auxiliary-source studies with different prediction targets, shared encoders follow the same update rule, while each source-specific head is aggregated only over laboratories using that prediction target. Checkpoint selection uses the original target's development panels. Each source retains its own prediction objective while contributing updates to the shared representation.

### 3.3 Proposal revision from development evidence

The first proposal is generated once without development feedback and shared by direct and loop search. History feedback begins with the second proposal. For subsequent research steps $t$, let $a_t$ be the retained design, $\mathcal{E}_t$ the development evidence accumulated so far and $\mathcal{U}_t$ the untried configurations. The research model $S$, with weights $\theta$ held fixed, generates a proposal $p_t$ and the coordinator maps its `design_id`, denoted $\mathrm{id}(p_t)$, to the candidate:

$$p_t \sim P_\theta(\cdot \mid \mathcal{T}, a_t, \mathcal{E}_t, \mathcal{U}_t), \qquad \widetilde{a}_t = \mathrm{Compile}(\mathrm{id}(p_t)). \tag{3}$$

The proposal also contains a `hypothesis`, an `experiment` and an `expected_effect`, recording the motivation, intended comparison and anticipated outcome. The identifier determines execution; measured development performance determines selection.

After training $\widetilde{a}_t$, the coordinator constructs the next card $e_{t+1}$. Comparing $q$ lexicographically by higher score and then lower loss gives

$$a_{t+1} = \begin{cases} \widetilde{a}_t, & q(\widetilde{a}_t) >_{\mathrm{lex}} q(a_t), \\ a_t, & \text{otherwise}, \end{cases} \qquad \mathcal{E}_{t+1} = \mathcal{E}_t \parallel [e_{t+1}], \tag{4}$$

where $\parallel$ appends a card to the history. Unsuccessful trials remain in the history. Invalid proposals and numerically failed fits consume a proposal slot and leave the incumbent unchanged. Updated context carries these outcomes into subsequent proposals. After the proposal budget is exhausted, the retained design and its selected checkpoint are evaluated on the held-out task.

### 3.4 Evidence-guided training allocation

We use trajectory-guided allocation in a controlled study (Section 4.4) to test whether early evaluation evidence improves the use of a fixed training budget. The screening–promotion policy retains both high-scoring and still-improving candidates. It screens ten predetermined designs for 20 rounds, evaluating every five rounds through Equation 2. Within each of three learning-rate groups, it promotes the best-scoring design and, among the remaining designs, the one with the largest score improvement from round 5 to round 20. Development loss breaks score ties. The six promoted designs restart from their initial parameters for 100-round training, and their best development checkpoints determine the final predictor.

The two promotion criteria reserve training capacity for candidates at different stages of convergence. Current score favors a strong screening checkpoint, while improvement over the shared screening interval retains a candidate whose score is still rising. Grouping by learning rate preserves representation from each group in the full-training stage. All promotion decisions use development measurements.

The control trains all ten designs for 80 rounds. Both schedules use 800 aggregation rounds and 160 aggregate development evaluations. This study evaluates allocation over a predetermined candidate set independently of proposal generation. The main comparison and proposal-history experiments use 100 training rounds per candidate. Section 4.4 reports the allocation results; Appendix A provides the complete proposal schema, algorithm and allocation rules.

# 4. Experiments

## 4.1 Tasks, datasets and evaluation

We evaluate BioCoLoop on three biological prediction problems with task-specific base models and a common procedure for collaborative training, evaluation and proposal generation. The main setting partitions each dataset into ten disjoint groups. Complementary studies use distinct biological datasets or experimental scenarios as laboratories to evaluate source transfer.

**Drug–target interactions.** Given a compound and a protein sequence, the model predicts whether they interact. BindingDB provides 50,149 training pairs and 5,604 development pairs, with canonical compounds defining laboratory assignments. We use TAPB as the base model, with frozen ESM2 protein features and trainable drug, fusion and classification modules (Lin et al., 2025). The primary metric is AUROC on 5,505 random test pairs. Additional test sets containing 3,791 unseen-drug pairs and 2,507 unseen-protein pairs measure generalization to unseen entities. AP, MCC, accuracy and cross-entropy are reported as secondary metrics.

**Observed-response proteomics.** Given a drug representation and its measured 6h and 24h proteomic responses, the model predicts binary efficacy. A ProteinTalks PTPC subset provides 386 training, 105 development and 148 test conditions (Sun et al., 2026). Compound-group assignment gives each laboratory 49-50 training and development conditions. The base model is a ProteinTalks-derived efficacy model with protein and drug branches, trained from random initialization. Average precision (AP) is the primary metric. AUROC, accuracy and cross-entropy are secondary metrics. Unlike de novo response prediction, this task conditions on observed post-treatment proteomic measurements.

**Cell perturbation identification.** Given an expression response and five candidate interventions, the model identifies the intervention whose predicted response has the highest cosine similarity to the observation. We use VCC single-gene perturbations (Arc Institute, 2025), Norman double-gene perturbations (Norman et al., 2019) and a Tahoe subset of drug perturbations in A549 at 5 $\mu$M (Zhang et al., 2025). Intervention-disjoint training, development and test counts are 60/20/20, 70/20/27 and 180/60/67, respectively. The base model uses a frozen scDEBART backbone with a trainable response head (Sung & Kim, 2026). All methods use the same corrected attention masks and the same candidate sets. The primary metric is intervention-macro Top-1, which first averages accuracy within each intervention and then across interventions. Random selection has an expected Top-1 of 20%. Micro Top-1, MRR, Top-3 and cross-entropy are reported as secondary metrics. Appendix A provides the response construction, chemical conditioning, training losses, and secondary evaluation details.

## 4.2 Compared methods and shared protocol

We compare BioCoLoop with three types of baselines. The *fixed model* fits the task-specific base model once with a predefined configuration. *Direct optimization* uses BioCoLoop's proposal generator and registered executable design space, generating proposals without previous development outcomes. AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) are adapted to the same biological tasks and fitting interface. Every method returns one predictor; laboratories contribute to its fitting, not an inference-time ensemble.

BioCoLoop combines collaborative fitting and evaluation with development-history feedback. We use Qwen2.5-7B-Instruct and GPT-5.6 Luna as proposal generators (OpenAI, 2026). Table 1 is an *access-scenario comparison*: BioCoLoop uses ten laboratories and the other methods use laboratory 0. It compares complete workflows with different available data and local computation. The same-access experiments in Section 4.7 and Appendix E instead fix all ten training laboratories, development panels, design choices and candidate-training budgets to examine search itself. Here *direct* means feedback-free proposal generation, not a single training run; the fixed-model control is the single-fit reference.

The main comparison uses twelve configurations and at most six proposals beyond the initial design. Each valid candidate starts from the prescribed task initialization and receives 100 training rounds. Development score and loss determine the retained design and checkpoint before held-out scoring. We report means and sample standard deviations over seeds 42–44 on fixed data partitions.

Table 1: Access-scenario comparison: ten participating laboratories versus one. Scores $\times 100$ are mean $\pm$ sample SD over seeds 42–44. Data and local computation differ with access; this table measures complete workflows, not the isolated effect of proposal history. Bold and underline indicate the best and second-best complete scores within each block. †: two scored seeds; –: no sealed score; ⋆: extended-context adaptation. Execution outcomes are in Appendix C; same-access search is reported separately.

| Method | Labs | DTI | Proteomics | Cell perturbation (Top-1) | | |
|---|---|---|---|---|---|---|
| | | AUROC ↑ | AP ↑ | VCC ↑ | Norman ↑ | Tahoe ↑ |
| **Qwen2.5-7B-Instruct** | | | | | | |
| Fixed model | 1 | 82.83 ± 1.22 | <u>25.05</u> ± 4.56 | <u>29.94</u> ± 0.00 | **25.56** ± 0.00 | <u>19.40</u> ± 0.00 |
| Direct | 1 | <u>89.89</u> ± 1.02 | 20.13 ± 0.07 | 21.09 ± 4.38 | 14.15 ± 0.00 | 15.42 ± 6.21 |
| AI-Scientist-v2 | 1 | 89.68 ± 0.80 | 20.18 ± 0.43 | 18.61 ± 2.78 | 14.12 ± 0.04 | 15.42 ± 6.21 |
| AI-Researcher | 1 | 86.09 ± 0.88 | 20.12 ± 0.05 | 19.70 ± 2.87† | 21.73 ± 6.63 | 17.41 ± 6.21 |
| **BioCoLoop** | 10 | **93.76** ± 1.11 | **36.99** ± 0.22 | **31.54** ± 0.08 | <u>22.25</u> ± 0.45 | **22.39** ± 2.59 |
| **GPT-5.6 Luna (low reasoning)** | | | | | | |
| Fixed model | 1 | 82.83 ± 1.22 | <u>25.05</u> ± 4.56 | <u>29.94</u> ± 0.00 | **25.56** ± 0.00 | <u>19.40</u> ± 0.00 |
| Direct | 1 | <u>89.89</u> ± 1.02 | 20.17 ± 0.42 | 18.61 ± 2.78 | 21.75 ± 6.59 | 15.92 ± 6.03 |
| AI-Scientist-v2 | 1 | 89.59 ± 0.95 | 23.37 ± 5.89 | 18.61 ± 2.78 | 14.12 ± 0.04 | 11.94 ± 2.11† |
| AI-Researcher | 1 | – | 20.29 ± 0.27⋆ | 18.61 ± 2.78⋆ | 14.12 ± 0.04⋆ | 15.42 ± 6.21⋆ |
| **BioCoLoop** | 10 | **93.67** ± 0.98 | **37.11** ± 0.01 | **31.50** ± 0.00 | <u>22.69</u> ± 0.73 | **22.39** ± 2.59 |

These summaries describe training and search variability; paired cluster-bootstrap intervals in the Statistical Supplement quantify held-out sample uncertainty. The allocation, source-transfer and extended-search studies use their separately specified budgets and seeds.

### 4.3 Performance with broader laboratory participation

Table 1 compares complete workflows under the specified access scenario. BioCoLoop improves four of five primary endpoints over the single-laboratory fixed model in both proposal-model blocks. The fixed model retains the highest Norman Top-1. The explicit laboratory-count column distinguishes these access gains from the same-access search effects in Section 4.7.

With Qwen, BioCoLoop reaches 93.76 AUROC on BindingDB and 36.99 AP on proteomics; with Luna, it reaches 93.67 and 37.11, respectively. The corresponding fixed-model scores are 82.83 and 25.05. BioCoLoop also attains the highest complete VCC and Tahoe scores in both blocks, at 31.54/31.50 and 22.39 macro Top-1. On Norman, the fixed model scores 25.56, compared with 22.25 for Qwen BioCoLoop and 22.69 for Luna BioCoLoop.

Single-laboratory random search, Bayesian optimization and twelve-design enumeration also remain below the ten-laboratory Qwen workflow (Appendix D). They test whether extra search compensates for limited data access; the ten-laboratory controls in Appendix E address search quality. Appendix E.7 additionally measures fixed-model variation across all ten single-laboratory identities.

### 4.4 Evidence-guided training allocation

Evidence-guided allocation improves held-out performance at a matched budget (Table 2). Uniform allocation trains all ten designs for 80 rounds. The evidence-guided policy screens them for 20 rounds and restarts six selected designs for 100-round training. Each schedule uses 800 aggregation rounds and 160 development evaluations.

Across twelve task–seed comparisons, allocation yields four wins, eight ties and no losses. Norman Top-1 increases from 11.48% to 22.32%, a 10.84-point gain with a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. A selected low-learning-rate model improves beyond the uniform stopping point. Tahoe improves in one seed and ties in two; proteomics and VCC tie in all three. Screening and restarting costs are included. This is an allocation result over a fixed candidate set, independent of language-model proposal history.

A fresh 360-fit comparison reproduces these means and adds simple promotion controls (Appendix E.5). On Norman, score-only promotion reaches 21.70 and the exact random-promotion expectation

Table 2: Evidence-guided training allocation at a matched budget. Held-out scores use the same ten designs, ten laboratories, 800 full-client aggregation rounds and 160 aggregate development evaluations per method. Uniform allocation gives every design 80 rounds; the evidence-guided scheduler screens all designs for 20 rounds and restarts six promoted designs for 100-round training. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [−1.00, 7.46]. The separate DTI development replay is in Appendix A.

| Benchmark | Uniform | Evidence-guided | Gain | W/T/L |
|---|---|---|---|---|
| Proteomics: efficacy | **37.47** | **37.47** | +0.00 | 0/3/0 |
| Cell: VCC single-gene | **20.01** | **20.01** | +0.00 | 0/3/0 |
| Cell: Norman double-gene | 11.48 | **22.32** | **+10.84** | 3/0/0 |
| Cell: Tahoe drug | 20.90 | **22.89** | **+1.99** | 1/2/0 |

is 22.63, versus 22.32 for score-plus-trajectory promotion. On Tahoe, the trajectory policy reaches 22.89 versus 20.90 and 21.95, respectively. Random promotion also outperforms the trajectory policy on VCC (22.49 versus 20.01). Thus selective longer fitting improves on uniform allocation, but the trajectory criterion does not consistently outperform simpler promotion rules.

### 4.5 Learning from different data sources

Source-transfer studies use laboratories from distinct biological datasets or experimental settings. We combine BindingDB with BioSNAP, Davis and Human; VCC with three external single-cell studies; and PTPC with Lin, Ruprecht and decryptE (Lin et al., 2026; Ruprecht et al., 2020; Eckert et al., 2025). Target development data select each model, and all configurations use the same target test set.

Transfer gains depend on the source. Adding all three proteomic studies improves target AP, and Davis and Human improve DTI AUROC. With laboratory and training-condition counts matched, external single-cell studies yield lower target Top-1 than additional data from VCC. Appendix B reports the source-specific effects and matched controls.

With one proteomic study per laboratory, the three auxiliary sources raise target AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39. Shared encoders combine local updates while source-specific heads retain their own objectives. In the matched cell study, both arms use four laboratories and 60 training interventions: same-source VCC reaches 30.70% Top-1 versus 28.82% with external studies. Appendix B details these source-defined simulations; Appendix E adds three-seed direct/history controls with the same source access.

### 4.6 Effect of the number of laboratories

We vary $K \in \{1, 2, 5, 10\}$ by either adding laboratories and training data or repartitioning a fixed total data pool. Adding data through participation can improve performance, while fixed-pool repartitioning has task-dependent effects (Figure 3). Proteomics performs best at an intermediate $K$, and Norman degrades as the fixed pool is split across more laboratories. Performance depends on both data access and the partition used for local training.

Both settings use the default design and a fixed training schedule. Increasing participation adds the corresponding local development panels, while fixed-pool repartitioning preserves equal-weight evaluation over all ten original panels. For the cell tasks, local training groups also define the contrastive-negative pool, so partitioning affects the learning objective's available negatives as well as the sequence of updates. Appendix B.1 reports the two settings separately with their corresponding evaluation panels.

### 4.7 Same-access configuration search

Table 3 fixes ten-laboratory access and adds random search without replacement and Gaussian-process expected improvement (GP–EI) to the paired direct/history study. The retrospective comparison reuses all twelve complete fits per task. Development-only selection is sealed before scoring; random expectations integrate all 462 six-design subsets. Appendix E gives search-policy variability, shared-first-proposal controls and sensitivities matching realized fit counts when proposals fail.

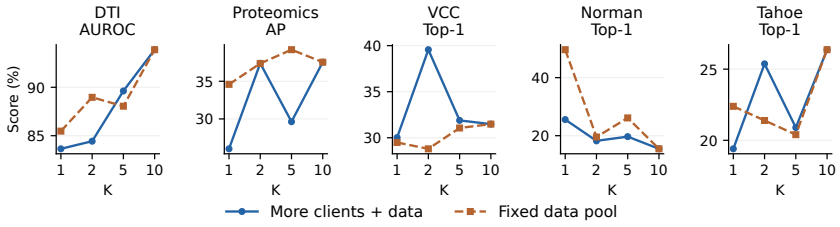



Figure 3: Effect of the number of laboratories. Solid lines add both laboratories and training data. Dashed lines divide the same training data across different numbers of laboratories. Full results are reported in Appendix B.

Table 3: Same-access search: all methods use ten laboratories and the same development panels. Searches use twelve designs, six proposal slots and at most 700 aggregation rounds, including initialization; the fixed reference uses 100 rounds. Scores $\times 100$ are conditional on one training seed per task (DTI 42; others 61), not the three-seed access comparison. Random is its exact policy expectation; variability and actual-fit sensitivities are in Appendix E. Bold marks the best unrounded score, including ties.

| Method | DTI AUROC ↑ | Proteomics AP ↑ | VCC Top-1 ↑ | Norman Top-1 ↑ | Tahoe Top-1 ↑ |
|---|---|---|---|---|---|
| Fixed model | 94.27 | **37.27** | **31.50** | 15.48 | 25.37 |
| Random, exact expectation | 94.45 | **37.27** | 27.53 | 22.66 | 26.59 |
| GP–EI | **94.60** | **37.27** | **31.50** | 22.30 | 25.37 |
| Qwen direct | 94.27 | **37.27** | **31.50** | 22.30 | 25.37 |
| Qwen history | 94.27 | **37.27** | **31.50** | 22.30 | 25.37 |
| Luna direct | **94.60** | **37.27** | **31.50** | 22.30 | **26.87** |
| Luna history | 94.27 | **37.27** | **31.50** | **23.11** | **26.87** |

On Norman, Luna history reaches 23.11 Top-1 versus 22.30 for direct/GP–EI and a random expectation of 22.66. On DTI, direct Luna and GP–EI instead outperform both history rows. Proteomics retains the fixed model; VCC's language-model approaches and GP–EI also match their fixed reference. Appendix B consolidates every short and extended comparison: history yields one gain, one decrease and eight ties in short search, and seven ties and one decrease in extended search. Retrospective target crossings describe search paths, not realized compute savings.

A fresh three-seed study enlarges cell development coverage to 414 perturbation genes across four source-defined contexts (Appendix E.6). Direct, structured history, random and GP–EI receive the same 16-design library, six requests and 16,940 optimizer steps per search. Structured history scores 22.89% held-out Top-1 versus 22.62% for direct and 22.67% for random; GP–EI scores 23.15%. The structured-minus-direct interval spans zero. These controls show a small positive point estimate in this setting, rather than consistent superiority over conventional search.

### 4.8 Validation under extended search

Re-evaluating 14 selected VCC/Tahoe checkpoints reproduces the original scores. Equal-laboratory and pooled-intervention macro scores coincide because panels contain equal numbers of interventions. Qwen/VCC nevertheless gains 1.17 development points and loses 11.89 test points; Tahoe gains 5.00 and loses 11.94. Under 20 paired development-option rosters, these advantages become $-1.34$ and $-1.33$ points on average, identifying sensitivity to the evaluation examples rather than an aggregation error. In the larger-data study, every final search outcome exceeds its fitted reference, although local development/test reversals remain. Appendix E reports all prefixes without test-based stopping or reselection.

## 5. Conclusion

BioCoLoop connects laboratory-local model fitting to iterative configuration evaluation through a common biological research interface. Its experiments distinguish broader data access, proposal history and training allocation. Compatible external studies can improve a target model, and selective

longer fitting benefits double-gene perturbation identification relative to uniform allocation. Simple-promotion controls show that this gain is not unique to the trajectory rule. Same-access search and development-panel diagnostics identify where a richer experiment history alone is insufficient. The resulting framework makes these decisions separately measurable and provides a basis for improving collaborative model development through better validation and allocation.

## Reproducibility statement

Section 3 specifies the research and training interfaces; Section 4 defines task inputs, partitions, reference models and metrics. Appendix A gives the proposal schema, design library, optimization settings and allocation rules. Appendices B and C document source/sensitivity studies and public-agent adaptations; D and E separate single-laboratory search from same-access search, allocation and validation studies. The Statistical Supplement reports complete original per-seed contrasts, including non-positive differences. The research records retain code and data hashes, seeds, development trajectories, selection decisions and saved predictions. The original proposal log covers all 504 attempted slots ($20 \times 6$ short-protocol and $16 \times 24$ extended-protocol slots). The revision adds retrospective cache-based search controls, checkpoint diagnostics, and a separate 360-fit allocation comparison, with their own selection seals, source bindings and independent score checks. Random-search variability is distinguished from training-seed variability. Qwen generation is seed-reproducible; Luna and public-agent interface constraints are specified in Appendices A and C.

The fresh history-representation study in Appendix E.6 adds 48 newly fitted candidates and three paired Qwen search repetitions. Its released summaries include all methods, exact random-policy expectations, step/example accounting and every development-selected prefix. The extension archive records source hashes and protocol versions; the earlier parser diagnostic is retained separately from the final scientific comparison.

## Ethics statement

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. Raw data remain local in the simulated laboratory protocol. Multi-institutional deployment would require a privacy and security policy for parameter updates and aggregate diagnostics. Predictions are evaluated for research use and have no biological or therapeutic validation.

## AI use statement

The main comparison retains matched proposal records for local Qwen2.5-7B-Instruct and the service identified as `gpt-5.6-luna`. Luna is run at low reasoning effort through a text-only bridge with tools disabled. Its service does not expose a generation seed, so the seed labels identify the training/search repetitions rather than deterministic language-model sampling. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, and manuscript drafting and revision. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

## References

Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8): 1657–1661, 2025. doi: 10.1038/s41592-025-02772-6. URL https://doi.org/10.1038/s41592-025-02772-6.

Arc Institute. Virtual cell challenge. Arc Virtual Cell Atlas dataset, 2025. URL https://github.com/ArcInstitute/arc-virtual-cell-atlas/tree/4f7bb170c61d2fafce70f67db6e52c7d9af1be4b/virtual-cell-challenge. 2025 data release; training file adata_Training.h5ad. Documentation accessed September 22, 2026.

Mohammad Bakhtiari, Stefan Bonn, Fabian Theis, Olga Zolotareva, and Jan Baumbach. FedscGen: privacy-preserving federated batch effect correction of single-cell RNA sequencing data. *Genome Biology*, 26:216, 2025. doi: 10.1186/s13059-025-03684-6. URL https://doi.org/10.1186/s13059-025-03684-6.

James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13(10):281–305, 2012. URL https://jmlr.org/papers/v13/bergstra12a.html.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11(70):2079–2107, 2010. URL https://jmlr.org/papers/v11/cawley10a.html.

Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480, 2024. doi: 10.1038/s41592-024-02201-0. URL https://doi.org/10.1038/s41592-024-02201-0.

Ittai Dayan, Holger R. Roth, Aoxiao Zhong, Ahmed Harouni, Amilcare Gentili, Anas Z. Abidin, Andrew Liu, Anthony Beardsworth Costa, et al. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine*, 27(10):1735–1743, 2021. doi: 10.1038/s41591-021-01506-3. URL https://doi.org/10.1038/s41591-021-01506-3.

Jiayuan Ding, Jianhui Lin, Ziyang Miao, Nils Mechtel, Shiyu Jiang, Yixin Wang, Zhaoyu Fang, Jorge D. Martin-Rufino, Chen Weng, Reuben Saunders, Weize Xu, Jonathan S. Weissman, Min Li, Jiliang Tang, Wei Ouyang, Yuancheng Ryan Lu, and Xiaojie Qiu. Predictive single cell foundation model for gene regulation and aging with privacy-preserving tabular learning, 2026. URL https://arxiv.org/abs/2607.19400v1. Version 1.

Stephan Eckert, Nicola Berner, Karl Kramer, Annika Schneider, Julian Müller, Severin Lechner, Sarah Brajkovic, Amirhossein Sakhteman, Christian Graetz, Jonas Fackler, Michael Dudek, Michael W. Pfaffl, Percy Knolle, Stephanie Wilhelm, and Bernhard Kuster. Decrypting the molecular basis of cellular drug phenotypes by dose-resolved expression proteomics. *Nature Biotechnology*, 43:406–415, 2025. doi: 10.1038/s41587-024-02218-y. URL https://doi.org/10.1038/s41587-024-02218-y. First published online 7 May 2024.

GAIR-NLP DrugEvolve contributors. DrugEvolve: Official source and applications, 2026. URL https://github.com/GAIR-NLP/DrugEvolve/tree/e945c7d0da28ddce73904419116a51e03921e6e6. Commit e945c7d0da28ddce73904419116a51e03921e6e6.

Shanghua Gao, Ada Fang, and Marinka Zitnik. AutoScientists: Self-organizing agent teams for long-running scientific experimentation, 2026. URL https://arxiv.org/abs/2605.28655v1. Version 1, 27 May 2026.

HKUDS. AI-Researcher: Official source, 2025. URL https://github.com/HKUDS/AI-Researcher/tree/f9a6f8480860c193afff600eeffe3defcee8a978. Commit f9a6f8480860c193afff600eeffe3defcee8a978.

Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. MLAgentBench: Evaluating language agents on machine learning experimentation, 2023. URL https://arxiv.org/abs/2310.03302v1. Version 1, 5 October 2023.

Longda Jiang, Carol Dalgarno, Efthymia Papalexi, Isabella Mascio, Hans-Hermann Wessels, Huiyoung Yun, Nika Iremadze, Gila Lithwick-Yanai, Doron Lipson, and Rahul Satija. Systematic reconstruction of molecular pathway signatures using scalable single-cell perturbation screens. *Nature Cell Biology*, 27:505–517, 2025a. doi: 10.1038/s41556-025-01622-z. URL https://doi.org/10.1038/s41556-025-01622-z.

Zhengyao Jiang, Dominik Schmidt, Dhruv Srikanth, Dixing Xu, Ian Kaplan, Deniss Jacenko, and Yuxiang Wu. AIDE: AI-driven exploration in the space of code, 2025b. URL https://arxiv.org/abs/2502.13138v1. Version 1, 18 February 2025.

Peter Kairouz, H. Brendan McMahan, et al. Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021. doi: 10.1561/2200000083. URL https://doi.org/10.1561/2200000083.

Georgios A. Kaissis, Marcus R. Makowski, Daniel Rückert, and Rickmer F. Braren. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence*, 2 (6):305–311, 2020. doi: 10.1038/s42256-020-0186-1. URL https://doi.org/10.1038/s42256-020-0186-1.

Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank J. Reddi, Sebastian U. Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning, 2020. URL https://arxiv.org/abs/1910.06378v1. Version 1, 14 October 2019; published at ICML 2020.

Mikhail Khodak, Renbo Tu, Tian Li, Liam Li, Maria-Florina Balcan, Virginia Smith, and Ameet Talwalkar. Federated hyperparameter tuning: Challenges, baselines, and connections to weight-sharing. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://papers.neurips.cc/paper_files/paper/2021/hash/a0205b87490c847182672e8d371e9948-Abstract.html.

Haoyuan Li, Mathias Funk, and Aaqib Saeed. Helmsman: Autonomous synthesis of federated learning systems via collaborative LLM agents. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=Voiy13SK3r. Method details also available in arXiv:2510.14512v2.

Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks, 2020. URL https://arxiv.org/abs/1812.06127v1. Version 1, 14 December 2018; published at MLSys 2020.

Chuwei Lin, Catherine M. Sniezek, Christopher D. McGann, Rashmi Karki, Ross M. Giglio, Benjamin A. Garcia, José L. McFaline-Figueroa, and Devin K. Schweppe. Defining the heterogeneous molecular landscape of lung cancer cell responses to epigenetic inhibition. *Communications Biology*, 9:176, 2026. doi: 10.1038/s42003-025-09455-0. URL https://doi.org/10.1038/s42003-025-09455-0. Published 28 January 2026.

Gaoming Lin, Xin Zhang, Zhonghao Ren, Quan Zou, Prayag Tiwari, Changjun Zhou, and Yijie Ding. TAPB: an interventional debiasing framework for alleviating target prior bias in drug-target interaction prediction. *Nature Communications*, 16:10867, 2025. doi: 10.1038/s41467-025-66915-1. URL https://doi.org/10.1038/s41467-025-66915-1.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery, 2024. URL https://arxiv.org/abs/2408.06292v1. Version 1, 12 August 2024.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative refinement with self-feedback, 2023. URL https://arxiv.org/abs/2303.17651v1. Version 1, 30 March 2023.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pp. 1273–1282, 2017. URL https://proceedings.mlr.press/v54/mcmahan17a.html.

Nils Mechtel, Wei Ouyang, and Chiron Platform contributors. Chiron Platform: Federated single-cell training and agent interfaces, 2026. URL https://github.com/aristoteleo/chiron/tree/3241b35c8b516c126aa68da450708de3da717ee0. Commit 3241b35c8b516c126aa68da450708de3da717ee0.

Ajay Nadig, Joseph M. Replogle, Angela N. Pogson, Mukundh Murthy, Steven A. McCarroll, Jonathan S. Weissman, Elise B. Robinson, and Luke J. O'Connor. Transcriptome-wide analysis of differential expression in perturbation atlases. *Nature Genetics*, 57:1228–1237, 2025. doi: 10.1038/s41588-025-02169-3. URL https://doi.org/10.1038/s41588-025-02169-3.

Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, Alex Y. Ge, Albert Xu, Marco Jost, Luke A. Gilbert, and Jonathan S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365(6455):786–793, 2019. doi: 10.1126/science.aax4438. URL https://doi.org/10.1126/science.aax4438.

OpenAI. The GPT-5.6 model family: the `gpt-5.6-luna` service, 2026. Proprietary reasoning service accessed through the OpenAI Codex CLI at low reasoning effort with tools disabled; the service exposes no generation seed. Accessed September 2026.

Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, Jeffrey A. Hussmann, Alexander Lenail, Alina Guna, Lauren Mascibroda, Eric J. Wagner, Karen Adelman, Gila Lithwick-Yanai, Nika Iremadze, Florian Oberstrass, Doron Lipson, Jessica L. Bonnar, Marco Jost, Thomas M. Norman, and Jonathan S. Weissman. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.e28, 2022. doi: 10.1016/j.cell.2022.05.013. URL https://doi.org/10.1016/j.cell.2022.05.013.

Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletarì, Holger R. Roth, Shadi Albarqouni, Spyridon Bakas, Mathieu N. Galtier, Bennett A. Landman, Klaus Maier-Hein, Sébastien Ourselin, Micah Sheller, Ronald M. Summers, Andrew Trask, Daguang Xu, Maximilian Baust, and M. Jorge Cardoso. The future of digital health with federated learning. *npj Digital Medicine*, 3:119, 2020. doi: 10.1038/s41746-020-00323-1. URL https://doi.org/10.1038/s41746-020-00323-1.

Yusuf Roohani, Andrew Lee, Qian Huang, Jian Vora, Zachary Steinhart, Kexin Huang, Alexander Marson, Percy Liang, and Jure Leskovec. BioDiscoveryAgent: An AI agent for designing genetic perturbation experiments, 2024. URL https://arxiv.org/abs/2405.17631v1. Version 1, 27 May 2024.

Benjamin Ruprecht, Julie Di Bernardo, Zhao Wang, Xuan Mo, Oleg Ursu, Matthew Christopher, Rafael B. Fernandez, Li Zheng, Brian D. Dill, Huijun Wang, Yuting Xu, Andy Liaw, Jonathan D. Mortison, Nirodhini Siriwardana, Brian Andresen, Meir Glick, James R. Tata, Victoria Kutilek, Ivan Cornella-Taracido, and An Chi. A mass spectrometry-based proteome map of drug action in lung cancer cell lines. *Nature Chemical Biology*, 16:1111–1119, 2020. doi: 10.1038/s41589-020-0572-3. URL https://doi.org/10.1038/s41589-020-0572-3.

SakanaAI. The AI scientist-v2: Official source, 2025. URL https://github.com/SakanaAI/AI-Scientist-v2/tree/96bd51617cfdbb494a9fc283af00fe090edfae48. Commit 96bd51617cfdbb494a9fc283af00fe090edfae48.

Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, Mikhail Milchenko, Weilin Xu, Daniel Marcus, Rivka R. Colen, and Spyridon Bakas. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10:12598, 2020. doi: 10.1038/s41598-020-69250-1. URL https://doi.org/10.1038/s41598-020-69250-1.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning, 2023. URL https://arxiv.org/abs/2303.11366v1. Version 1, 20 March 2023.

Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. Practical bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems*, 2012. URL https://arxiv.org/abs/1206.2944.

Rui Sun, Liujia Qian, Yongge Li, Tong Liu, Honghan Cheng, Xuedong Zhang, Xueya Zhou, Yuecheng Zhan, Guangmei Zhang, Zhengchao Luo, Kunpeng Ma, Chunlong Wu, Dongchen Ji, Zhangzhi Xue, Hongxue Meng, Yuhang Xiang, Dingwei Lei, Qianhe Zhou, Wenbin Hu, Yuhan Deng, Lingling Tan, Qi Xiao, Zhiwei Liu, Lei Zeng, Liqin Qian, Xuan Zheng, Qiong Hu, Nianzi

Luo, Weinan E, Peijie Zhou, Han Wen, Yi Zhu, and Tiannan Guo. An operational perturbation proteomics-based virtual cell model. *Nature*, 2026. doi: 10.1038/s41586-026-11001-9. URL https://doi.org/10.1038/s41586-026-11001-9. Published online 9 September 2026.

Jieun Sung and Wankyu Kim. scDEBART: Predicting in silico single-cell perturbation responses via large-scale differential expression learning. In *International Conference on Machine Learning*, 2026. URL https://openreview.net/forum?id=pJyidZg93y.

Jiabin Tang, Lianghao Xia, Zhonghang Li, and Chao Huang. AI-Researcher: Autonomous scientific innovation, 2025. URL https://arxiv.org/abs/2505.18705v1. Version 1, 24 May 2025.

Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, Helene Mantineo, Elizabeth M. Brydon, Zexian Zeng, X. Shirley Liu, and Patrick T. Ellinor. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023. doi: 10.1038/s41586-023-06139-9. URL https://doi.org/10.1038/s41586-023-06139-9.

Shuang Wang, Bochen Shen, Lanting Guo, Mengqi Shang, Jinze Liu, Qi Sun, and Bairong Shen. scFed: federated learning for cell type classification with scRNA-seq. *Briefings in Bioinformatics*, 25(1):bbad507, 2024. doi: 10.1093/bib/bbad507. URL https://doi.org/10.1093/bib/bbad507.

Yutaro Yamada, Robert Tjarko Lange, Cong Lu, Shengran Hu, Chris Lu, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist-v2: Workshop-level automated scientific discovery via agentic tree search, 2025. URL https://arxiv.org/abs/2504.08066v1. Version 1, 10 April 2025.

Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2309.03409.

Harrison G. Zhang, Peter Eckmann, Jiacheng Miao, Andrew B. Mahon, and James Zou. The virtual biotech: A multi-agent AI framework for therapeutic discovery and development. *Science*, 394 (6822):eaeg6779, 2026. doi: 10.1126/science.aeg6779. URL https://doi.org/10.1126/science.aeg6779.

Jesse Zhang, Airol A. Ubas, Richard de Borja, Valentine Svensson, Nicole Thomas, Neha Thakar, Aidan Winters, Umair Khan, Matthew G. Jones, John D. Thompson, Vuong Tran, Joseph Pangallo, Efthymia Papalexi, Ajay Sapre, Hoai Nguyen, Oliver Sanderson, Maria Nigos, Olivia Kaplan, Sarah Schroeder, Bryan Hariadi, Simone Marrujo, Crina Curca Alec Salvino, Guillermo Gallareta Olivares, Ryan Koehler, Gary Geiss, Alexander Rosenberg, Charles Roco, Daniele Merico, Nima Alidoust, Hani Goodarzi, and Johnny Yu. Tahoe-100M: A giga-scale single-cell perturbation atlas for context-dependent gene function and cellular modeling. *bioRxiv*, 2025. doi: 10.1101/2025.02.20.639398. URL https://www.biorxiv.org/content/10.1101/2025.02.20.639398v3. Version 3, May 10, 2025.

Zhimeng Zhou, Yang Nan, Minjie Mou, Yuntao Qian, Yixiu Liu, Zhengyu Zuo, Hao Yang, Weixian Xu, Bo Li, Wanghao Jiang, Yanlin Ren, Yang Liao, Yimeng Wang, Yinghong Li, Qingxia Yang, Zhiheng Xi, Tiantian Mi, Huaicheng Sun, Pengfei Liu, and Feng Zhu. An AI system for autonomous algorithm evolution in drug development. *bioRxiv*, 2026. doi: 10.64898/2026.08.16.745117. URL https://doi.org/10.64898/2026.08.16.745117. Version 1, posted 20 August 2026.

# Appendix A. Algorithm and task protocols

**How to read the supplementary experiments.** Appendix A specifies the shared algorithm and task scorers. Appendix B separates laboratory count, source transfer and proposal-history studies; Appendix C documents task-adapted public agents. Appendix D compares search strategies restricted to one laboratory. Appendix E holds laboratory access fixed when comparing search and training-promotion policies, and examines development–test divergence. These studies answer different questions: a ten-versus-one laboratory gain measures the complete access scenario, whereas same-access comparisons isolate the research decision.

| Study | What changes | What the comparison measures |
|---|---|---|
| Access scenario | One versus ten laboratories and research workflow | Combined data-access and workflow effect |
| Same-access search | Proposal policy; common data and menu | Configuration-search quality |
| Source transfer | Biological study supplying local updates | Positive or negative transfer to the target |
| Training allocation | Which candidates receive longer fitting | Use of a fixed training/evaluation budget |
| Roster diagnostic | Development distractors and option order | Sensitivity of already selected models |

## A.1 Research algorithm and optimization

All three task families use the parameter update, development-selection rule and proposal interface of Section 3. A task adapter defines the input representation, prediction components, training loss and development scorer. The research process below starts from the task's reference design. The fitted reference also supplies its first evidence card.

| | **BioCoLoop research procedure** |
|---|---|
| 1 | Fit the reference design from the task initialization and seed, then retain its best development checkpoint and initialize the research history. |
| 2 | Generate the shared first proposal without development feedback. From the second slot onward, supply the task, untried configurations, retained design and prior evidence cards. |
| 3 | Parse the proposal and map its `design_id` to an executable configuration. Allow at most two generation attempts in the slot. |
| 4 | Fit a valid candidate from the same initialization procedure and seed for the assigned number of aggregation rounds. |
| 5 | Select its best development checkpoint. Replace the incumbent only if its score/loss pair is better. |
| 6 | Append the executed configuration, hypothesis, diagnostics and decision to the history, including rejected or failed trials. |
| 7 | After the final slot, return the retained design, its selected checkpoint and the trial history, and score that checkpoint on the held-out task. |

In the controlled direct/loop comparison, the first proposal is generated once without development feedback and shared. History enters loop search from the second proposal. Direct search subsequently receives the default configuration and untried identifiers without development outcomes; history-guided search receives the retained configuration and accumulated cards. Both use the same registered executable design space, proposal count and candidate-training schedule. Appendix B reports every prespecified proposal prefix. A configuration evaluated previously under the same task, data access and seed can reuse its completed fit while retaining its logical position and budget charge in each search.

Each aggregation round contains one local AdamW epoch per participating laboratory. The batch size is 64, gradient-norm clipping is one, and the local optimizer state resets each round. The learning rate and weight decay follow the selected design. The proximal penalty is zero throughout the reported design libraries. The coordinator weights local parameter states by their number of training examples. Coordinator momentum follows Equation 1 and applies only to trainable parameters. Floating-point buffers use sample-weighted averaging, while integer and Boolean buffers use an elementwise maximum. These buffers store non-parameter model state. Every candidate starts with the same task-specific public or random initialization procedure and study seed. The optional residual module has width 32 and a zero-initialized output, while the original prediction head remains trainable. The implementation calls each laboratory a *client* or *worker*, and its configuration field

`server_momentum` stores the coordinator momentum coefficient. These names refer to the same laboratory and coordinator roles used in Section 3.

Full-training candidates receive 100 aggregation rounds, with development evaluation every five rounds. The selection key is mean primary score followed by negative mean loss over the development panels designated by the protocol (Section 3.2). In the main experiment these panels are the participating laboratories. A complete tie retains the earlier checkpoint or incumbent. Training-example counts refer to the adapter's training rows: for the cell tasks these are intervention-level training profiles, rather than the number of cells underlying a profile.

### A.2 Registered executable design space, proposals and evidence

The prefix `D` stands for *design*; its two-digit suffix identifies a configuration. D00 is the starting configuration. The main study selects six further proposals from D01–D11 without repeating a design within a search. The table maps each identifier to its four-factor configuration, with a zero-initialized width-32 residual module when enabled. Appendix B specifies the extended design space and its identifiers.

| Design | Learning rate | Weight decay | Momentum | Residual module |
|---|---|---|---|---|
| D00 | 0.001 | 0.001 | 0 | no |
| D01 | 0.0001 | 0.001 | 0 | no |
| D02 | 0.01 | 0.001 | 0 | no |
| D03 | 0.001 | 0 | 0 | no |
| D04 | 0.001 | 0.01 | 0 | no |
| D05 | 0.001 | 0.001 | 0.9 | no |
| D06 | 0.001 | 0.001 | 0 | yes |
| D07 | 0.001 | 0.001 | 0.9 | yes |
| D08 | 0.0001 | 0.001 | 0 | yes |
| D09 | 0.0001 | 0.001 | 0.9 | yes |
| D10 | 0.01 | 0.01 | 0 | no |
| D11 | 0.001 | 0.01 | 0 | yes |

The main comparison has two proposal-model blocks. Qwen2.5-7B-Instruct uses bfloat16 inference, temperature 0.7, top-$p$ 0.9 and at most 512 new tokens. For zero-indexed proposal slot $t$ and generation attempt $b \in \{0, 1\}$, its sampling seed is $z + t + 10{,}000b$, where $z$ is the study seed. GPT-5.6 Luna uses a text-only bridge at low reasoning effort with tools disabled and the same bounded proposal protocol. Its service exposes no generation seed; the seed labels identify training and search repetitions. An invalid identifier, repeated configuration or malformed response triggers the second attempt. A second failure consumes the slot without fitting a candidate. A numerically failed fit leaves the incumbent unchanged and remains in the evidence history.

A proposal has exactly four fields:

```
{
  "hypothesis": "why the proposed change may help",
  "experiment": "the intended controlled comparison",
  "expected_effect": "the anticipated metric or diagnostic change",
  "design_id": "D08"
}
```

The three descriptive fields must be nonempty strings and record the agent's rationale. The identifier must belong to the registered executable design space and determines the executed configuration.

An evidence card contains the executed configuration, its hypothesis, the best development score and loss, the selected round, and the acceptance decision. Its trajectory summary includes the first and last development-evaluation rounds, their sample-weighted training losses, equal-laboratory development scores and losses, and the last-minus-first changes in development score and loss. For a 100-round fit these endpoints are rounds 5 and 100. In the main comparison, the development summary also records the worst laboratory score and the number of evaluating laboratories. Training participation and the designated evaluation panels are specified separately for source-transfer and repartitioning studies. Ordered cards are supplied as context, and unsuccessful trials are not removed.

**An observed proposal sequence.** For Norman with ten laboratories and seed 42, D06 is accepted in the first slot: it ties D00's development Top-1 of 40.00 but reduces development loss from 158.98 to 158.94. In the second slot, D08 retains the residual module and changes the learning rate from $10^{-3}$ to $10^{-4}$. Its development Top-1 rises from 10.20 at round 5 to 51.90 at round 100. Its development loss falls from 168.12 to 153.85. The candidate is accepted. The next proposed design, D07, reaches a best Top-1 of 45.00 and is rejected. Thus the history records both the slower-improving accepted candidate and the later unsuccessful alternative.

Three representative trajectories are summarized below. Scores and losses are multiplied by 100 and proposal slots are one-indexed.

| Study | Selected design | Primary | Loss | First direct / loop slot |
|---|---|---|---|---|
| PTPC, seed 42 | D04 | 66.58 | 52.12 | 5 / 5 |
| VCC, seed 43 | D00 | 24.04 | 160.78 | initial baseline |
| Norman, seed 42 | D08 | 51.90 | 153.85 | 3 / 2 |

PTPC selects stronger weight decay through the loss tie-break at equal AP, while VCC retains its initial design. Section 4.7 compares search trajectories under the shared design space, proposal budget and selection rule.

### A.3 Independent training-allocation control

This study fixes the candidate set and changes only the allocation of training rounds. It uses seeds 53–55 on proteomics and the three cell settings. The ten configurations cover the following factor combinations. Every configuration uses the same ten laboratories.

| Learning rate | Weight decay | Momentum | Residual module |
|---|---|---|---|
| 0.001 | 0.001 | 0 | no |
| 0.0001 | 0.001 | 0 | no |
| 0.0001 | 0.001 | 0 | yes |
| 0.0001 | 0.01 | 0.9 | no |
| 0.0001 | 0.01 | 0.9 | yes |
| 0.001 | 0.001 | 0.9 | yes |
| 0.01 | 0.001 | 0 | yes |
| 0.01 | 0.001 | 0.9 | yes |
| 0.01 | 0.01 | 0.9 | no |
| 0.001 | 0.01 | 0 | yes |

Uniform allocation trains every design for 80 rounds. Trajectory-guided allocation first trains each for 20 rounds. Within each of the three learning-rate groups, it promotes the design with the highest primary score at its best screening checkpoint, using lower development loss to break ties. From the remaining designs in that group, it also promotes the largest development-score improvement from round 5 to round 20, using lower round-20 development loss to break ties. Complete ties follow the order of the configurations in the table. The six promoted designs restart from their initial parameters and train for 100 rounds. Final selection considers their full-training checkpoints, while the four unpromoted designs stop after screening.

Both schedules use 800 aggregation rounds: $10 \times 80 = 10 \times 20 + 6 \times 100$. With ten participating laboratories, this is 8,000 local epochs. Evaluation every five rounds gives 160 aggregate development evaluations, each formed from ten laboratory evaluations. These are matched training and evaluation counts. The scheduler itself generates no language-model proposals.

Across the 12 task–seed held-out comparisons, allocation gives four wins, eight ties and no losses. Norman improves by 10.84 Top-1 points on average, with seed-level effects of +10.00, +11.70 and +10.81 and a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. Tahoe gives one +5.97-point seed and two ties, and VCC and PTPC tie in all three seeds. A separate TAPB development replay uses the seven-design intersection of existing trajectories. Uniform allocation receives 525 candidate-rounds and trajectory-guided allocation receives 520, with a mean development AUROC difference of +0.39 points (two wins and one tie). It is not included in the twelve held-out comparisons.

### A.4 Data construction and task adapters

**Drug–target interaction.** The BindingDB partition contains 50,149 fitting and 5,604 development pairs. Canonical compounds are assigned to ten laboratories, giving 4,932–5,058 training and 517–643 development pairs per laboratory. The fixed test sets contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. TAPB uses frozen public protein features and trainable drug, fusion and classification components. An optional residual adapter precedes its native confounder-averaged prediction. Classification NLL and masked-language loss form its local training objective.

**Observed-response proteomics.** The PTPC source pool contains 333 compound groups and 491 conditions, partitioned into 386 fitting and 105 development conditions. A laboratory owns 35–42 fitting conditions. The test set contains 148 conditions. For each 6h and 24h profile, preprocessing applies log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is separately L2-normalized and scaled by 0.5. The two 5,171-dimensional proteomic blocks and drug vector form a 12,390-dimensional float32 input. The ProteinTalks-derived efficacy head has protein and drug convolutional branches followed by a 64–32–1 classifier, totaling 464,193 parameters under D00. It is trained from random initialization using binary cross-entropy. All comparison methods use these same inputs and architecture. AP is the selection metric.

**Cell inputs and response units.** VCC, Norman and Tahoe use whole-intervention fitting, development and test splits of 60/20/20, 70/20/27 and 180/60/67. Training uses one mean response profile per intervention. VCC development and test queries average disjoint groups of up to 64 cells, omitting a final group smaller than 16 cells. Norman queries are the released scVI-derived differential-expression profiles for each gene pair. Tahoe uses one aggregate response profile per compound at the selected cell context and dose. These are the units scored by the identification task.

The scDEBART backbone receives the shared untreated reference and candidate perturbation encoding. Its features are fixed while the response head is trained. The corrected feature extractor implements three attention masks as additive blocking masks. Its correction preserves the pretrained weights, gene panels, response targets, intervention partitions and candidate options. Genetic-task scoring excludes the union of directly perturbed genes. Tahoe concatenates a fixed 128-dimensional projection of a 2,048-bit Morgan fingerprint to the 768-dimensional backbone features, with the additional head-input weights initialized to zero. Its drug projection uses seed 1234. The training loss is response MSE plus 0.1 times a contrastive loss over interventions in the local training minibatch. The cosine-logit temperature is 0.1.

**Five-option queries and scoring.** For every development or test intervention, four distinct distractors are sampled uniformly without replacement from the *same* development or test intervention pool, excluding the true intervention. The true intervention is added and the five positions are shuffled. A single roster is fixed per intervention and reused for all of its response queries, all training seeds and all methods. Split construction and roster sampling use the preparation generator initialized with seed 20260918. Training does not use these five-option rosters, and its contrastive negatives come from the local training minibatch.

For each query, predicted candidate responses and the observed response are L2-normalized over the scoring genes. Candidates are ranked by their cosine similarity. Equal scores retain the fixed shuffled option order. Let $G$ be the number of evaluated interventions, $m_g$ the number of response queries for intervention $g$, and $r_{g,j}$ the rank of the true option for its $j$th query. The primary score is

$$\text{MacroTop1} = \frac{1}{G}\sum_{g=1}^{G}\frac{1}{m_g}\sum_{j=1}^{m_g}\mathbf{1}\{r_{g,j}=1\}.$$

Thus each intervention receives equal weight even when it contributes a different number of query profiles. Micro Top-1, mean reciprocal rank and Top-3 average over individual queries. The development loss is cross-entropy over the five cosine scores divided by temperature 0.1. Development selection first computes these scores within each laboratory and then averages laboratories equally.

**Identification diagnostics.** The reported frozen-scDEBART identification diagnostics span 7.5–31.6% macro Top-1 against a 20% chance level and 36.00–57.00 MRR against a 45.67% chance level,

Table A.1: Control comparison for the task-reference models. The fixed and feedback-free direct references train or search on a single site, so they are reported on one laboratory; BioCoLoop is reported at one and ten laboratories so that the effect of laboratory participation can be read under the same research loop. Entries are held-out mean ± sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

| | | DTI | Proteomics | Cell perturbation | | |
|---|---|---|---|---|---|---|
| Labs | Method | AUROC ↑ | AP ↑ | VCC Top-1 ↑ | Norman Top-1 ↑ | Tahoe Top-1 ↑ |
| 1 | Fixed model | 82.83±1.22 | 25.05±4.56 | 29.94±0.00 | **25.56**±0.00 | 19.40±0.00 |
| 1 | Direct | 89.89±1.02 | 20.13±0.07 | 21.09±4.38 | 14.15±0.00 | 15.42±6.21 |
| 1 | Our loop | 89.89±1.02 | 20.13±0.07 | 20.22±2.22 | 14.15±0.00 | 15.42±6.21 |
| 10 | Our loop | **93.76**±1.11 | **36.99**±0.22 | **31.54**±0.08 | 22.25±0.45 | **22.39**±2.59 |
| Completed seeds ($n/3$) | | 3/3 | 3/3 | 3/3 | 3/3 | 3/3 |

since a uniform five-option ranking has expected reciprocal rank $(1 + 1/2 + 1/3 + 1/4 + 1/5)/5$. The untrained public head scores 7.5% on Norman. True-option positions are near-uniform across the fixed rosters, and a prototype oracle built from observed held-out responses retrieves the true option at 94.5% on VCC and 100% on Norman.

### A.5 Complete comparisons and uncertainty

The following tables report the control comparison, secondary metrics and all three drug–target test sets. Bold marks the best displayed value within a task and metric, including ties. Higher is better except for binary cross-entropy and cell cross-entropy.

Table A.2: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44 and multiplied by 100. Bold indicates all tied maxima, or minima for loss.

| Method | Labs | Metric 1 | Metric 2 | Metric 3 | Metric 4 |
|---|---|---|---|---|---|
| **DTI** ($n = 3$) | | AUROC | AP | MCC | BCE |
| Fixed model | 1 | 82.83 | 81.57 | 49.95 | 52.37 |
| Direct | 1 | 89.89 | 87.77 | 65.71 | 61.02 |
| Our loop | 1 | 89.89 | 87.77 | 65.71 | 61.02 |
| Our loop | 10 | **93.76** | **92.73** | **73.71** | **36.59** |
| **Proteomics** ($n = 3$) | | AP | AUROC | BCE | Acc. |
| Fixed model | 1 | 25.05 | 65.49 | 42.65 | **85.14** |
| Direct | 1 | 20.13 | 61.80 | 44.34 | 81.76 |
| Our loop | 1 | 20.13 | 61.80 | 44.34 | 81.76 |
| Our loop | 10 | **36.99** | **73.87** | **39.83** | **85.14** |
| **VCC** ($n = 3$) | | Micro Top-1 | MRR | Top-3 | CE |
| Fixed model | 1 | 33.61 | **56.53** | **73.89** | **154.51** |
| Direct | 1 | 21.48 | 47.10 | 62.78 | 160.82 |
| Our loop | 1 | 20.74 | 46.94 | 64.26 | 160.86 |
| Our loop | 10 | **33.98** | 56.34 | 69.44 | 158.14 |
| **Norman** ($n = 3$) | | Micro Top-1 | MRR | Top-3 | CE |
| Fixed model | 1 | **25.56** | 48.50 | 63.93 | **158.74** |
| Direct | 1 | 14.15 | 42.12 | 58.35 | 160.39 |
| Our loop | 1 | 14.15 | 42.12 | 58.35 | 160.39 |
| Our loop | 10 | 22.25 | **50.07** | **70.96** | 158.83 |
| **Tahoe** ($n = 3$) | | Micro Top-1 | MRR | Top-3 | CE |
| Fixed model | 1 | 19.40 | 45.00 | 58.21 | 163.08 |
| Direct | 1 | 15.42 | 42.60 | 56.72 | 161.90 |
| Our loop | 1 | 15.42 | 42.60 | 56.72 | 161.90 |
| Our loop | 10 | **22.39** | **47.95** | **61.19** | **160.75** |

Table A.3: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified evaluation. The main DTI endpoint is random held-out.

| Endpoint | Method | Labs | AUROC | AP |
|---|---|---|---|---|
| Random | Fixed model | 1 | 82.83 | 81.57 |
| Random | Direct | 1 | 89.89 | 87.77 |
| Random | Our loop | 1 | 89.89 | 87.77 |
| Random | Our loop | 10 | **93.76** | **92.73** |
| Unseen drug | Fixed model | 1 | 81.96 | 85.78 |
| Unseen drug | Direct | 1 | 88.39 | 89.35 |
| Unseen drug | Our loop | 1 | 88.39 | 89.35 |
| Unseen drug | Our loop | 10 | **91.65** | **92.58** |
| Unseen protein | Fixed model | 1 | 76.40 | 78.75 |
| Unseen protein | Direct | 1 | 83.26 | 80.53 |
| Unseen protein | Our loop | 1 | 83.26 | 80.53 |
| Unseen protein | Our loop | 10 | **87.63** | **85.07** |

Three-seed means and sample standard deviations summarize training/search variability on fixed partitions. Sample-level intervals use paired cluster bootstrap with 1,000 fixed-seed resamples of

retained predictions. Drug identity defines clusters for the random and unseen-drug interaction sets, and protein sequence defines clusters for the unseen-protein set. Proteomic and cell intervals use compound and intervention clusters, respectively. Both methods in a contrast receive the same resampled clusters. The training-allocation study instead uses 5,000 hierarchical paired resamples: training seeds are drawn first, followed by compound or intervention groups within each sampled seed.

The accompanying *BioCoLoop Statistical Supplement*, Tables S.1–S.2, reports all per-seed differences and confidence intervals for the main contrasts and three drug–target test sets, including zero and negative differences. These model-conditional intervals quantify test-sample uncertainty; several cell comparisons include zero.

The accompanying research records contain the executable configurations, proposals, development histories and saved predictions needed to reconstruct these comparisons. The reproducibility statement describes their availability and verification.

### A.6 Scope and limitations

The experiments simulate multiple laboratories on shared compute using public data. They evaluate the research protocol without measuring communication or straggler costs or providing formal privacy guarantees. Model development is confined to the finite registered executable design space, with four editable factors and one optional residual module. Qwen generation is seed-reproducible; the proprietary Luna service exposes no generation seed. Public self-improving agents use task-adapted interfaces with the component restrictions and incomplete executions documented in Appendix C. The study includes no centralized pooled-data reference using the same complete data pool.

# Appendix B. Laboratory, source and proposer sensitivity

**Sensitivity study design.** We examine collaboration scale, source diversity, proposer choice and research budget using the same target scorers as the main study. Laboratory-count experiments cover all five endpoints at $K \in \{1, 2, 5, 10\}$ under both increasing participation and a fixed total data pool. DTI uses seed 61, while the lighter proteomics and cell experiments use seeds 61–63. Independent-source DTI and proteomics use seed 61, and external-source VCC uses seeds 61–63. The two-proposer study has two separate protocols: a 12-design, six-slot study on all five endpoints (DTI seed 42; other endpoints seed 61), and a 36-design, 24-slot study on the four lighter endpoints at seed 61. Each candidate receives 100 training rounds. The latter changes the design space as well as the search budget, so budget comparisons are made within a protocol. Main-table results retain their original seeds 42–44 and protocol.

## B.1 Number of laboratories

The participation experiment uses a fixed nested set of the first $K$ original laboratories, increasing both training and development access. The fixed-pool experiment merges the complete ten-laboratory pool into $K$ training groups while retaining equal-weight evaluation over the original ten development panels. Thus its selection population stays fixed as training partitions change. The identical $K = 10$ fit is reused by both comparisons. Every unique fit uses the default design and 100 rounds. The cell objective forms contrastive negatives within each local training group, so repartitioning changes the negative pool as well as optimization steps. These comparisons characterize the implemented training pipeline at different collaboration scales. Zero between-seed SD can occur when a fixed pretrained head receives an identical single-batch update, which describes training variability on these fixed partitions.

Table B.1: Increasing participation. A nested set of the original ten client partitions participates, so increasing $K$ also adds training data. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best $K$ within each row, including displayed ties.

| Task | Metric | $K = 1$ | $K = 2$ | $K = 5$ | $K = 10$ |
|---|---|---|---|---|---|
| DTI | AUROC | 83.65 | 84.43 | 89.62 | **93.88** |
| PTPC | AP | 26.03 ± 5.57 | 37.38 ± 0.60 | 29.63 ± 2.50 | **37.54 ± 0.27** |
| VCC | Macro Top-1 | 30.02 ± 0.78 | **39.56 ± 0.11** | 31.88 ± 0.00 | 31.48 ± 0.16 |
| Norman | Macro Top-1 | **25.56 ± 0.00** | 18.22 ± 0.00 | 19.68 ± 1.11 | 15.48 ± 0.00 |
| Tahoe | Macro Top-1 | 19.40 ± 0.00 | 25.37 ± 0.00 | 20.90 ± 0.00 | **26.37 ± 0.86** |

Table B.2: Fixed data pool. All target data are retained and merged into $K$ clients; only the partition count changes. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best $K$ within each row, including displayed ties.

| Task | Metric | $K = 1$ | $K = 2$ | $K = 5$ | $K = 10$ |
|---|---|---|---|---|---|
| DTI | AUROC | 85.47 | 88.96 | 88.05 | **93.88** |
| PTPC | AP | 34.59 ± 5.66 | 37.36 ± 1.89 | **39.19 ± 0.43** | 37.54 ± 0.27 |
| VCC | Macro Top-1 | 29.49 ± 0.00 | 28.80 ± 0.20 | 31.06 ± 0.22 | **31.48 ± 0.16** |
| Norman | Macro Top-1 | **49.63 ± 4.20** | 19.56 ± 0.20 | 26.12 ± 0.11 | 15.48 ± 0.00 |
| Tahoe | Macro Top-1 | 22.39 ± 0.00 | 21.39 ± 0.86 | 20.40 ± 0.86 | **26.37 ± 0.86** |

## B.2 Independent sources and target evaluation

For DTI and cell transfer, the entire target training pool forms one laboratory, and each auxiliary collection contributes a separate local update through sample-weighted aggregation. Table B.3 distinguishes these source-defined laboratories from the within-corpus partitions used in the laboratory-count study. Only the original target development panels select the checkpoint, and every configuration is evaluated on the same target test set. Target-only, individual-source and combined-source configurations start from the same task initialization.

DTI auxiliaries are BioSNAP, Davis and Human from the pinned TAPB release, with 19,092, 1,721 and 2,929 retained training pairs after drug–protein-pair de-duplication against target splits and pre-

Table B.3: Source and context mapping in the cross-source experiments. The DTI units are database collections, the cell units preserve their study and biological scenario, and each proteomic auxiliary study owns one client and its response head. The ten PTPC clients remain partitions of one target corpus. Biological contexts within a study are not counted as additional independent studies.

| Task | Client unit | Sources and biological contexts |
|---|---|---|
| DTI | Four data collections | Target BindingDB; BioSNAP; Davis; Human. Each collection contributes one client after pair-level de-duplication. |
| Cell | Four study/scenario clients | Target VCC; Replogle K562; Nadig HepG2; Jiang IFNB/K562. Each auxiliary source contributes one client. |
| Proteomics | Ten target partitions plus three source clients | Clients 0–9: PTPC partitions; client 10: Lin (A549, H292, PC9, PSC1); client 11: Ruprecht (A549, Calu6, Calu1, 2030, 2122); client 12: decryptE (Jurkat, five doses). |

ceding sources. All five DTI source configurations complete the same 100-round schedule. VCC auxiliary data comprise Replogle K562, Nadig HepG2 and Jiang IFNB/K562, with respectively 60, 60 and 58 retained training interventions (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Source feature panels preserve measured genes, and target development/test intervention identities are excluded from source training. The source manifest records retained subsets, feature intersections and exclusions.

Table B.4: Independent-source transfer on the same target heldout set, including every prespecified source and their combination. DTI (seed 61) adds BioSNAP, Davis or Human; VCC (seeds 61–63, mean ± SD) adds Replogle, Nadig or Jiang. The PTPC auxiliary multi-task model (seed 61) adds Lin or Ruprecht and is compared only with its matched auxiliary target-only model. The two-source PTPC protocol shown here uses Lin and Ruprecht; the three-source extension is reported separately in Table B.6. Changes are paired against target-only training; bold marks the best score within a target/model block. Scores are multiplied by 100.

| Target | Additional source | Metric | Score ↑ | Δ (pp) |
|---|---|---|---|---|
| DTI | Target only | AUROC | 85.47 | +0.00 |
| DTI | + BioSNAP | AUROC | 82.44 | -3.03 |
| DTI | + Davis | AUROC | 87.86 | +2.39 |
| DTI | + Human | AUROC | **87.95** | +2.48 |
| DTI | + All three | AUROC | 84.30 | -1.18 |
| VCC | Target only | Macro Top-1 | 29.49 ± 0.00 | +0.00 ± 0.00 |
| VCC | + Replogle | Macro Top-1 | 27.12 ± 0.08 | -2.37 ± 0.08 |
| VCC | + Nadig | Macro Top-1 | 26.94 ± 0.18 | -2.55 ± 0.18 |
| VCC | + Jiang | Macro Top-1 | **33.07 ± 0.00** | +3.58 ± 0.00 |
| VCC | + All three | Macro Top-1 | 27.33 ± 0.00 | -2.15 ± 0.00 |
| PTPC auxiliary | Target only | AP | 34.98 | +0.00 |
| PTPC auxiliary | + Lin | AP | 35.01 | +0.03 |
| PTPC auxiliary | + Ruprecht | AP | **35.50** | +0.52 |
| PTPC auxiliary | + Both | AP | 35.31 | +0.34 |

The cell study reports mean and SD across three seeds with a paired intervention-bootstrap analysis. Jiang improves target Top-1 by 3.58 points, with a 95% bootstrap interval of $[-2.11, 9.51]$. Source-specific gains and decreases are reported in Table B.4.

**Matched source–scenario laboratories.** Table B.5 compares four fixed-design settings: 15 VCC training interventions in one laboratory, all 60 VCC interventions in one laboratory, the same 60 partitioned into four laboratories, and four source-defined laboratories with 15 interventions each. The latter uses VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562. Both four-laboratory arms share the same 15-intervention VCC anchor. Hash-based sampling fixes the auxiliary identities before fitting, excluding target training, development and test intervention identities as well as duplicates across auxiliary laboratories. This experiment matches training intervention counts, not raw-cell counts: the same-source arm represents 77,771 cells, versus 33,469 in the cross-scenario arm. Each source retains its measured gene panel and corresponding loss mask.

All four settings use identical head initialization, the default design, 100 rounds and seeds 61–63. The original ten target-development panels select checkpoints, and the common target test contains 20

Table B.5: Fixed-design source–scenario ablation on VCC. $K$: training clients; $N$: training intervention conditions. The two $K = 4, N = 60$ arms share a 15-condition VCC anchor and differ in the sources of the other three clients. All arms use 100 rounds and the same target evaluation. Mean $\pm$ sample SD over three seeds; best means are bold. Cell counts and gene panels remain source-specific; protocol details are in Appendix B.

| Training sources | $K$ | $N$ | Top-1 (%) ↑ | MRR (%) ↑ |
|---|---|---|---|---|
| VCC, limited data | 1 | 15 | $29.36 \pm 0.26$ | $\mathbf{56.97 \pm 0.19}$ |
| VCC, all target data | 1 | 60 | $29.43 \pm 0.09$ | $56.00 \pm 0.09$ |
| VCC, same-source split | 4 | 60 | $\mathbf{30.70 \pm 0.42}$ | $56.24 \pm 0.04$ |
| VCC + external studies | 4 | 60 | $28.82 \pm 0.00$ | $56.30 \pm 0.00$ |

interventions and 360 five-option queries. No proposal search is performed. All twelve fits complete, and independent evaluation of saved predictions reproduces the metrics. Cross-scenario Top-1 is lower in each seed, with a mean difference of $-1.88$ points relative to the same-source four-laboratory setting; the MRR difference is near zero. The protocol and source identities remain fixed throughout the comparison.

**Two-source proteomic transfer and proposal search.** Two independent lung-cancer studies provide auxiliary measurements: Lin et al. (Lin et al., 2026) and Ruprecht et al. (Ruprecht et al., 2020). Their continuous response labels differ from the binary target endpoint. A shared masked-protein encoder of width 64 and drug encoder of width 32 therefore feed source-specific prediction heads. Target fitting uses its binary efficacy objective, Lin predicts continuous CellTiter-Glo response, and Ruprecht predicts an interval-valued potency outcome. Target data retain the original ten laboratories. Lin and Ruprecht each add one study-level laboratory. Shared encoders use sample-weighted aggregation, whereas each private head is updated only by its own source laboratories. The common laboratory and coordinator interfaces are reused. This adapter is compared against its own target-only control, separately from the main-table PTPC predictor.

The Lin subset contains 28 drug conditions (20 source-training, eight source-development), spanning five training and two development drugs. Inputs are 24-hour, 10-$\mu$M protein log-fold changes, and the response is the mean of the three released CellTiter-Glo ratio measurements. The HCT16/HCT116 alias is excluded from auxiliary training. The Ruprecht subset contains 109 matched drug conditions (85 training, 24 development; 17 and five drugs respectively). Protein log-fold changes at 24 hours are paired with CellTiter-Glo EC50 values in $\log_{10}$ nM: 49 exact, 18 approximate and 42 right-censored outcomes. Squared distance to the encoded potency interval is the loss. The adapter treats explicit lower bounds and values at or above 10,000 nM as right-censored and encodes approximate values with a factor-of-two interval. These are analysis conventions. Profiling dose is not a model input, although source acquisition uses dose conditions chosen in relation to drug response. The target retains 386 training and 105 development examples, with 148 held-out observations across 92 compounds and 22 positives.

All four source settings receive the same 100-round procedure. A two-proposal Luna loop is executed independently for each setting. These 12 fits and eight valid proposal calls produce eight fixed/loop evaluations, with identical held-out ranking scores within each source arm. The following four source-specific scores therefore describe both selection modes; this pilot measures source transfer without an additional history-search gain.

Target-only AP/AUROC are 34.98/74.75. Adding Lin yields 35.01/74.78, Ruprecht 35.50/74.28 and both 35.31/74.17. Paired compound bootstrap (1,000 resamples) gives 95% intervals for the AP changes of $[-0.20, 0.28]$, $[-1.46, 2.64]$ and $[-1.62, 2.49]$ points, respectively. These results measure transfer from two independent external studies and provide the source-matched controls for the two-proposal comparison.

**Three-source proteomic transfer.** The third auxiliary study, decryptE (Eckert et al., 2025), measures Jurkat proteomes after 18 hours of drug exposure at 1, 10, 100, 1,000 and 10,000 nM. Its source objective predicts the released continuous metabolic-activity measurement relative to DMSO. We align compounds using the published names, explicit alternative names and structures, and match protein features by UniProt identifier. Inputs comprise log-transformed measured normalized protein intensities, an observation mask, drug fingerprints and dose indicators. Splits group all doses of the

same chemical graph together, and normalization is fitted on source-training measurements. Target development and test compounds are excluded by canonical identity and stereo-agnostic chemical graph.

The repository's 145 main processed matrices contain one additional manually classified Vincristine copy. The uniform standard release therefore provides 144 compounds. Pre-training identity checks exclude 12 compounds overlapping the target development/test sets, six records with unsupported internal names and two records with conflicting released chemical-identity/structure annotations. The resulting source contains 124 distinct compounds and 620 dose conditions: 495 training observations from 99 compounds and 125 development observations from 25 compounds. The source manifest records every exclusion and preserves the original annotations.

We append a private continuous-response head to the shared adapter and instantiate this same four-head model in every control. All original encoder and head tensors retain identical seed-61 initialization. The four prespecified arms are target only, target plus decryptE, target plus Lin and Ruprecht, and target plus all three studies. Every arm completes 100 rounds with the same local optimizer and sample-weighted shared-parameter aggregation. Checkpoint selection uses equal-laboratory target-development AP with binary cross-entropy as the tiebreaker, and auxiliary development scores are diagnostic. These fixed-design comparisons use zero proposal slots and are separate from the preceding two-source research-loop study.

Table B.6: Three-source proteomic transfer on PTPC. All four arms use the same four-head adapter, seed 61, 100-round fitting schedule and target-development checkpoint rule. AP and AUROC are multiplied by 100; bold identifies the best value in each metric. The target test set contains 148 observations from 92 compounds. These are fixed-design source comparisons; the separate two-source proposal pilot is described in Appendix B.2.

| Additional source | AP ↑ | AUROC ↑ | Δ AP (pp) |
|---|---|---|---|
| Target only | 34.98 | **74.75** | +0.00 |
| + decryptE | 35.03 | 74.21 | +0.05 |
| + Lin + Ruprecht | 35.31 | 74.17 | +0.34 |
| + Lin + Ruprecht + decryptE | **35.68** | 74.28 | +0.70 |

Combining all three sources raises target AP from 34.98 to 35.68, a 0.70-point gain. Relative to Lin and Ruprecht together, adding decryptE increases AP by 0.37 points and AUROC by 0.11 points. Target-only training retains the highest AUROC across the four arms (74.75 versus 74.28 for all three). These source-transfer estimates use one seed. The four complete CPU fits take 101.21 seconds in total, and sealed checkpoints and predictions are independently rescored with exact agreement.

**One study per training laboratory.** To assign every training laboratory to a distinct source, we additionally merge all 386 target training examples into one laboratory. Lin, Ruprecht and decryptE each remain a separate source laboratory. The four arms use $K = 1, 2, 3, 4$ laboratories and respectively 386, 881, 491 and 986 training conditions. Their four-head architecture, seed-61 initialization, local loss, sample-weighted shared-parameter aggregation and source-private head ownership are unchanged. The merged target laboratory performs seven minibatches per local epoch, instead of aggregating updates from ten smaller target partitions. This is therefore a separately trained protocol, with a matching target-only control.

The original ten target-development panels still select checkpoints with equal panel weights, independently of the number of training laboratories. All four configurations complete 100 rounds and use zero proposal slots. Held-out AP and AUROC are independently recomputed from saved predictions, with maximum absolute discrepancy $1.11 \times 10^{-16}$. The full CPU run takes 76.18 seconds.

With one laboratory per study, all three auxiliary sources improve target AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39. Adding decryptE to Lin and Ruprecht increases AP by 1.44 points. These single-seed comparisons measure source transfer under a fixed design, varying available training data and participating studies.

**Within-study proteomic contexts.** HCC1143, HCC1806 and MDA-MB-453-1 are drawn from the mtPTDS supplementary collection (Sun et al., 2026), each with 47 retained drug conditions and 3,501 proteins shared with the target panel. They are three biological contexts within one study.

Table B.7: One worker per proteomic study. The target training pool is merged into one client; Lin, Ruprecht and decryptE each form one additional client. $K$ counts training clients and $N$ counts training conditions. All four fixed-design arms use 100 rounds, the same initialization (seed 61), and equal-weight evaluation over the original ten target development panels. AP and AUROC are percentages on the common target held-out set; best values are bold.

| Training sources | $K$ | $N$ | AP ↑ | AUROC ↑ |
|---|---|---|---|---|
| Target study | 1 | 386 | 34.41 | 72.33 |
| + decryptE | 2 | 881 | 36.58 | 73.81 |
| + Lin + Ruprecht | 3 | 491 | 36.09 | 72.69 |
| + Lin + Ruprecht + decryptE | 4 | 986 | **37.53** | **74.39** |

Target development/test compounds are excluded from source training. These experiments use the original proteomic adapter and seeds 61–63.

Table B.8: Within-study PTPC context transfer, reported separately from independent-source transfer. HCC1143, HCC1806 and MDA-MB-453-1 belong to the same mtPTDS study. Values are mean ± sample SD over seeds 61–63, multiplied by 100. Confidence intervals resample target compounds while retaining paired seed predictions. These contexts are not additional independent studies; the original model and results are unchanged.

| Additional context | AP ↑ | Δ AP (pp) | Paired 95% CI |
|---|---|---|---|
| Target only | 34.59 ± 5.66 | +0.00 ± 0.00 | [0.00, 0.00] |
| + HCC1143 | 34.94 ± 5.15 | +0.35 ± 0.56 | [-1.22, 1.60] |
| + HCC1806 | 35.23 ± 5.35 | +0.64 ± 0.32 | [-0.78, 1.77] |
| + MDA-MB-453-1 | 35.20 ± 5.31 | +0.61 ± 0.44 | [-0.85, 2.04] |
| + All three | **35.47 ± 6.36** | +0.87 ± 8.89 | [-2.44, 3.99] |

### B.3 Research-model and six-slot proposal comparison

The short study uses the 12 configurations in the registered executable design space of Appendix A. Direct search receives untried identifiers without measured scores, while the loop additionally receives accumulated development evidence. Both modes share initialization and the first proposal. Each slot allows two attempts to return the four-field proposal schema, and failed slots consume budget and remain in the trace. Every valid candidate restarts from the common task/seed initialization and is fitted for 100 rounds. Prefixes at 0, 1, 2, 4 and 6 slots are evaluated after the trajectory is complete. An exact fit cache can reuse an identical configuration while retaining its logical training-budget charge.

Qwen uses the local sampling settings in Appendix A. The second proposer is the `gpt-5.6-luna` service, accessed in a text-only session with tool use disabled, with request metadata and responses retained. The comparison holds task messages, editable designs, training procedure and proposal budget fixed across backends. Service-level decoding controls and parameter count are not exposed by the Luna interface.

All five endpoints have both backends and both search modes. Among the ten direct/loop pairs, Norman/Luna improves by 0.81 Top-1 points, DTI/Luna decreases by 0.33 AUROC points and eight pairs tie. The twenty DTI prefix scores were independently recomputed from saved predictions for all 5,505 test pairs, with checkpoint and output hashes verified. Figure 4 shows the complete prespecified prefix sequence for every endpoint, alongside its development selection trajectory.

### B.4 Extended search and task-dependent saturation

The extended study expands the registered executable design space to 36 configurations: three learning rates, three weight decays, two coordinator-momentum values and two residual-head settings. It runs 24 proposal slots for both backends and modes on PTPC, VCC, Norman and Tahoe. Prefixes are 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24. Initialization, full candidate fitting, development selection and invalid-slot accounting are unchanged. DTI's backend comparison is reported in the six-slot study in B.3.

Table B.9: Original and extended proposal searches, shown together without mixing configuration libraries. D and H denote direct and history-guided search, and @ denotes the number of proposal slots. Every candidate receives 100 training rounds using the same ten laboratories. Panel (a) uses DTI seed 42 and other-task seed 61; panel (b) uses seed 61. The six-slot prefixes in the two panels belong to different searches. DTI uses AUROC, proteomics uses average precision, and cell tasks use Top-1, all multiplied by 100. Bold denotes the highest displayed score in each row, including ties; dashes indicate runs not performed.

| Task | Proposer | Fixed | D@6 | H@6 | D@24 | H@24 |
|---|---|---|---|---|---|---|
| *(a) 12 designs, six proposal slots* | | | | | | |
| DTI | Luna | 94.27 | **94.60** | 94.27 | – | – |
| DTI | Qwen | **94.27** | **94.27** | **94.27** | – | – |
| Proteomics | Luna | **37.27** | **37.27** | **37.27** | – | – |
| Proteomics | Qwen | **37.27** | **37.27** | **37.27** | – | – |
| VCC | Luna | **31.50** | **31.50** | **31.50** | – | – |
| VCC | Qwen | **31.50** | **31.50** | **31.50** | – | – |
| Norman | Luna | 15.48 | 22.30 | **23.11** | – | – |
| Norman | Qwen | 15.48 | **22.30** | **22.30** | – | – |
| Tahoe | Luna | 25.37 | **26.87** | **26.87** | – | – |
| Tahoe | Qwen | **25.37** | **25.37** | **25.37** | – | – |
| *(b) 36 designs, 24 proposal slots* | | | | | | |
| Proteomics | Luna | **37.27** | **37.27** | **37.27** | **37.27** | **37.27** |
| Proteomics | Qwen | **37.27** | **37.27** | **37.27** | **37.27** | **37.27** |
| VCC | Luna | **31.50** | 16.94 | 16.94 | 20.23 | 20.01 |
| VCC | Qwen | **31.50** | **31.50** | **31.50** | 19.61 | 19.61 |
| Norman | Luna | 15.48 | **35.48** | 23.11 | 23.11 | 23.11 |
| Norman | Qwen | 15.48 | **22.30** | **22.30** | **22.30** | **22.30** |
| Tahoe | Luna | 25.37 | 13.43 | **26.87** | 13.43 | 13.43 |
| Tahoe | Qwen | 25.37 | **26.87** | **26.87** | 13.43 | 13.43 |

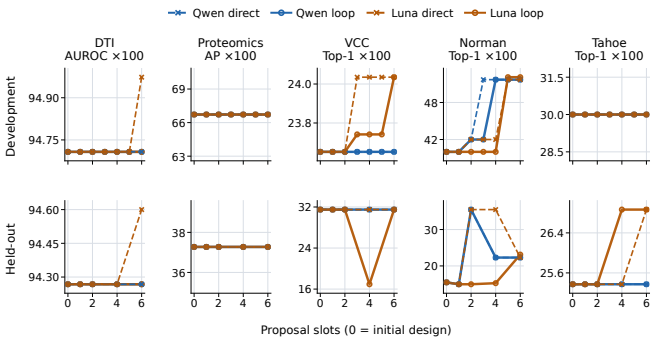



Figure 4: Research trajectories with and without evidence history in the six-slot protocol. Curves report development selection and held-out scores at each proposal prefix, where each candidate receives 100 rounds and slot 0 is the initial design.

Table B.10: Retrospective development-trajectory diagnostics for the two proposal-search protocols. Each D/H pair gives direct/history. The target is the lower terminal development score of the two runs; First is its earliest attainment, with slot 0 denoting initialization. Last gain records the final strict primary-score increase; Last retain also includes loss-tiebreak selections. All six or 24 slots were executed: these target crossings are descriptive and did not save training budget. Valid counts report accepted proposal formats, not retained model improvements.

| Task | Proposer | Dev. target | First D/H | Last gain D/H | Last retain D/H | Valid D/H |
|---|---|---|---|---|---|---|
| *(a) 12 designs, six proposal slots* | | | | | | |
| DTI | Luna | 94.71 | 0/0 | 6/0 | 6/0 | 6/6 |
| DTI | Qwen | 94.71 | 0/0 | 0/0 | 0/0 | 6/6 |
| Proteomics | Luna | 66.72 | 0/0 | 0/0 | 0/0 | 6/6 |
| Proteomics | Qwen | 66.72 | 0/0 | 0/0 | 0/0 | 6/5 |
| VCC | Luna | 24.04 | 3/6 | 3/6 | 3/6 | 6/6 |
| VCC | Qwen | 23.65 | 0/0 | 0/0 | 0/0 | 6/6 |
| Norman | Luna | 51.70 | 5/5 | 5/5 | 5/5 | 6/6 |
| Norman | Qwen | 51.70 | 3/4 | 3/4 | 3/4 | 6/6 |
| Tahoe | Luna | 30.00 | 0/0 | 0/0 | 6/3 | 6/6 |
| Tahoe | Qwen | 30.00 | 0/0 | 0/0 | 0/0 | 4/4 |
| *(b) 36 designs, 24 proposal slots* | | | | | | |
| Proteomics | Luna | 66.72 | 0/0 | 0/0 | 0/0 | 24/24 |
| Proteomics | Qwen | 66.72 | 0/0 | 0/0 | 0/0 | 19/14 |
| VCC | Luna | 27.68 | 16/9 | 16/10 | 16/10 | 24/24 |
| VCC | Qwen | 24.83 | 18/23 | 18/23 | 18/23 | 11/16 |
| Norman | Luna | 52.10 | 22/4 | 22/4 | 22/7 | 24/24 |
| Norman | Qwen | 51.70 | 2/5 | 2/5 | 22/19 | 15/12 |
| Tahoe | Luna | 35.00 | 5/13 | 5/13 | 5/13 | 24/24 |
| Tahoe | Qwen | 35.00 | 18/14 | 18/14 | 18/14 | 14/15 |

Luna produces valid designs in all 192 attempted slots, and Qwen in 116 of 192. Across 16 mode trajectories, the primary development score is unchanged in the final eight slots in 12 cases, and the full score/loss selection key is unchanged in ten. Each trajectory retains 11–24 unvisited designs, so these plateaus occur before the registered executable design space is exhausted.

Table B.10 measures the first slot attaining the smaller terminal development score within each direct/loop pair, a common target reached by both modes. This target is defined after both trajectories finish: it describes their paths, not an operational stopping rule. All 24 slots were executed, so an earlier crossing does not measure saved compute. Luna reaches Norman's common target earlier with feedback (4 versus 22 slots), whereas direct search reaches Tahoe's target earlier (5 versus 13). Proteomics retains the initial design. Table B.12 connects Norman's trajectory to concrete configuration changes: reducing the learning rate to $10^{-4}$ gives an early development gain, and removing the residual head reaches the final primary-score target. Subsequent loss-tiebreak selections can change the retained design without increasing that primary score.

Within this extended protocol, Qwen's VCC held-out Top-1 decreases from 31.50 at slot 6 to 19.61 at slot 24 while development selection improves. The development and held-out curves report this divergence alongside proposal validity and the remaining unvisited configurations.

### B.5 Executed designs and proposal traces

The short protocol uses the D00–D11 configurations in Appendix A.2. The extended registered executable design space contains all $3 \times 3 \times 2 \times 2 = 36$ combinations in Table B.11. An identifier L$i$W$j$M$k$R$l$ selects the indexed value of each factor, with $i, j \in \{0, 1, 2\}$ and $k, l \in \{0, 1\}$. For example, L0W1M0R1 specifies learning rate $10^{-4}$, weight decay $10^{-3}$, zero coordinator momentum and an enabled residual module. Every configuration has proximal coefficient zero, and the task-specific prediction interface remains fixed.

The reproducibility archive retains all 504 attempted proposal slots and their acceptance decisions: 120 slots in the short protocol and 384 in the extended protocol. Its machine-readable trace and

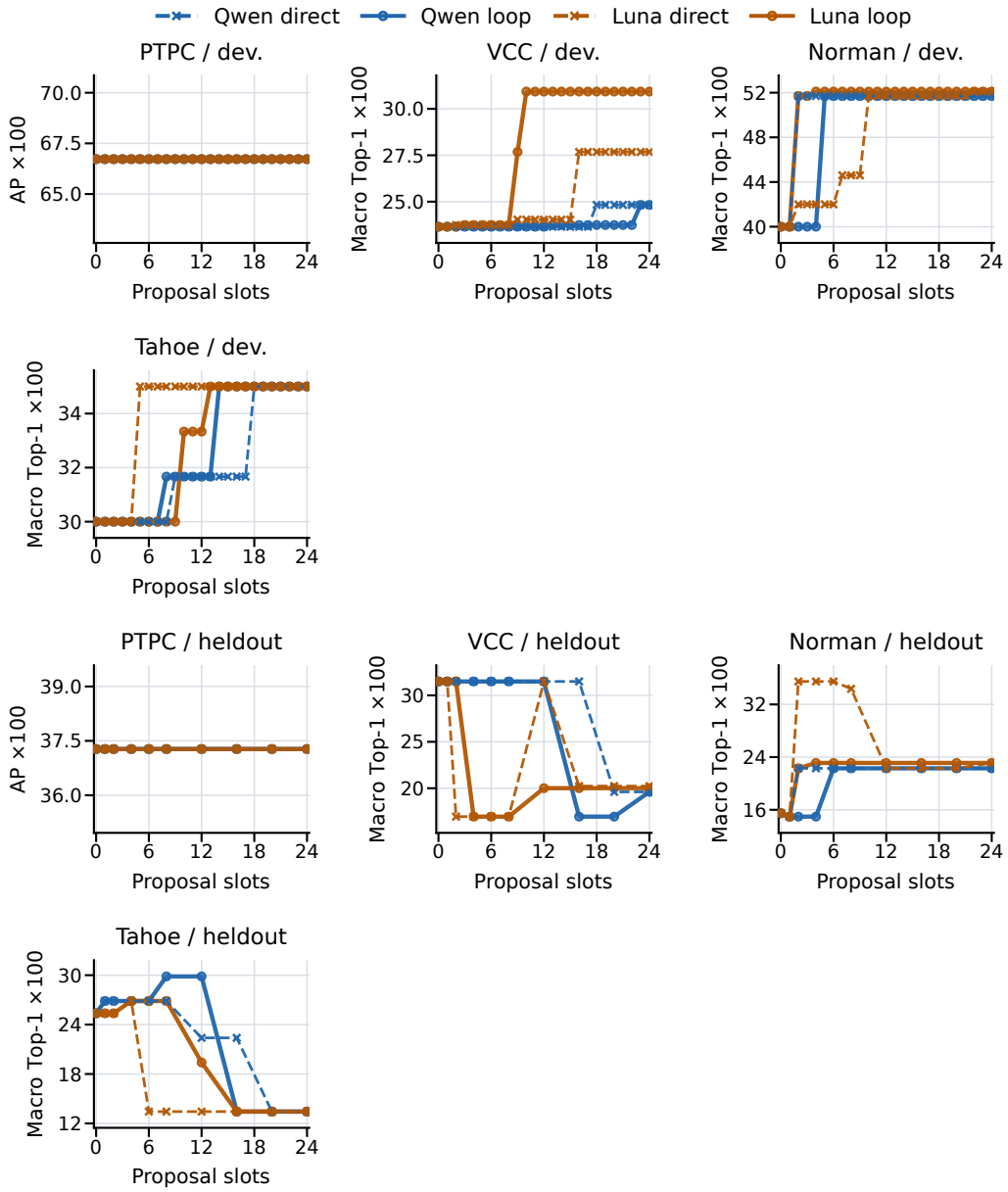



Figure 5: Development selection (upper panels) and held-out performance (lower panels) in the separate 36-design, 24-slot protocol on four endpoints. Curves are paired within the same backend, initialization and design space. Score plateaus may still include loss-tiebreak selections. All prespecified prefixes are displayed. Six-slot points here belong to this 36-design protocol, not the separate 12-design experiment.

Table B.11: Exact factor-to-index mapping for the extended 36-design library. Identifiers concatenate the four prefix–index pairs in L, W, M, R order, and all combinations of the listed values are included.

| Factor | Identifier prefix | Index 0 | Index 1 | Index 2 |
|---|---|---|---|---|
| Learning rate | `L` | $10^{-4}$ | $10^{-3}$ | $10^{-2}$ |
| Weight decay | `W` | 0 | $10^{-3}$ | $10^{-2}$ |
| Coordinator momentum | `M` | 0 | 0.9 | – |
| Residual module | `R` | disabled | enabled | – |

expanded tables record each proposal's configuration, changes from the incumbent and measured development score.

Table B.12: All retained proposals for the Norman/Luna extended 36-design, 24-slot study, seed 61. The two modes share initialization and first proposal. lr, wd, m and R denote learning rate, weight decay, server momentum and residual-head indicator. Development selection is followed by evaluation at every prespecified heldout prefix; the full attempted-slot trace is retained separately.

| Mode | Slot | Executed configuration | Dev. Top-1 |
|---|---|---|---|
| Direct | 0 | Common initial recipe | 40.00 |
| Direct | 1 | lr=0.001, wd=0.001, m=0, R=1 | 40.00 |
| Direct | 2 | lr=0.001, wd=0.001, m=0.9, R=1 | 42.00 |
| Direct | 7 | lr=0.001, wd=0.01, m=0.9, R=1 | 44.60 |
| Direct | 10 | lr=0.0001, wd=0, m=0, R=1 | 51.70 |
| Direct | 11 | lr=0.0001, wd=0.001, m=0, R=1 | 51.70 |
| Direct | 22 | lr=0.0001, wd=0.001, m=0, R=0 | 52.10 |
| Loop | 0 | Common initial recipe | 40.00 |
| Loop | 1 | lr=0.001, wd=0.001, m=0, R=1 | 40.00 |
| Loop | 2 | lr=0.0001, wd=0.001, m=0, R=1 | 51.70 |
| Loop | 4 | lr=0.0001, wd=0.001, m=0, R=0 | 52.10 |
| Loop | 7 | lr=0.0001, wd=0, m=0, R=0 | 52.10 |

# Appendix C. Public self-improving-agent comparison

## C.1 Shared biological task interface

AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) use BioCoLoop's fitting and evaluation service, at source versions `96bd51617cfd` and `f9a6f8480860`, respectively (SakanaAI, 2025; HKUDS, 2025). Both use the registered executable design space of twelve configurations, at most six proposals, 100 rounds per valid candidate and the shared held-out scorer. The service fits D00 afresh, then executes submitted `design_id` values; the hypothesis, experiment and expected-effect fields record rationale. Repeated or invalid requests consume slots, and agents may stop early. The original development score/loss rule selects checkpoints.

AI-Scientist-v2 retains its experiment manager, tree-search paths, reviews and journal. AI-Researcher retains native idea selection, planning, implementation, review handoffs and iterative analysis. Prompts use the registered design interface; arbitrary generated-code execution, external retrieval, figure review and manuscript generation are disabled. The matched blocks use local Qwen2.5-7B-Instruct (temperature 0.7, top-$p$ 0.9, at most 512 generated tokens per call) and GPT-5.6 Luna (low reasoning effort, text-only bridge, tools disabled). Adapters allow at most 160 model calls and 28,000 input tokens per call, raised to 65,536 for the four starred Luna AI-Researcher retries (C.2). Qwen uses generation seeds; Luna labels identify training/search repetitions. Per-run call counts, token usage and transport receipts document the variable language-model costs.

## C.2 Prediction quality and execution outcomes

A scored run completes its agent's research process within the registered cap, seals development selection and passes independent held-out rescoring. Table 1 summarizes seeds 42–44; fixed, direct and BioCoLoop rows are complete. Daggers mark partial cells and dashes mark unscored cells, without imputation. Partial cells are Qwen2.5-7B-Instruct AI-Researcher on VCC ($2/3$ seeds, $F_{\text{tool}}$) and GPT-5.6 Luna AI-Scientist-v2 on Tahoe ($2/3$, $F_{\text{svc}}$).

Four Luna AI-Researcher cells exceeded the 28,000-token admission ($F_{\text{ctx}}$) and were re-run at 65,536 tokens with the same tasks, seeds, design space, six-proposal cap, training and scorer. Their completed cells carry stars, with per-seed records retained. BindingDB (DTI), the longest native refinement trajectory, remains unscored ($0/3$): one frozen seed reached 29,167 input tokens ($F_{\text{ctx}}$), and two terminated in service transport ($F_{\text{svc}}$). Extended-admission retries also ended in transport before held-out selection was sealed. The revised table reports laboratory access instead of averaging ranks over different completed endpoint sets.

Receipts distinguish native pipeline ($F_{\text{pipe}}$), tool-transport ($F_{\text{tool}}$), registered context-limit ($F_{\text{ctx}}$) and Luna service-transport ($F_{\text{svc}}$) terminations before sealed selection. An earlier seed-42 Qwen AI-Scientist-v2 DTI attempt trained three candidates before substage schema validation failed. That historical attempt is distinct from the completed run represented in the table. Execution failures describe this adaptation and infrastructure, not predictive scores or evidence that the original framework cannot solve the task.

Held-out scoring follows sealed development selection. An independent verifier recomputes each scored cell's metric from saved predictions and checks checkpoint and artifact hashes. DTI discrepancies are at most $10^{-15}$. Records retain raw calls, native stage history, candidate requests, selections and verification receipts.

## C.3 Task-transport repair

The AI-Researcher adaptation removes repeated task packets from stage prompts, preserving the complete task and evidence in system prompts, native reasoning and tool results. The bridge accepts an unambiguous `parameters` alias for `arguments` and standard single-tool-call envelopes without inferring actions or values. This revision preserves model weights, context allowance, training and scoring. It applies across all five endpoints and the Luna sensitivity; failures remain recorded.

# Appendix D. Single-laboratory configuration-search controls

## D.1 Search protocol and primary results

We add conventional configuration-search controls to test whether more extensive search within one laboratory can match the collaborative framework. All new controls access only laboratory 0's training and development partitions. They use the same twelve registered designs, task-specific initialization, original GPU training implementation and held-out scorers as the main study. Each design is fitted afresh for 100 rounds at seeds 42–44. Development score, followed by development loss, selects the design and checkpoint.

Random search evaluates D00 and six distinct randomly sampled designs. Bayesian optimization evaluates D00, the first two proposals from the corresponding seeded random-search sequence, and four proposals selected by Gaussian-process expected improvement. Its fixed Matérn-$5/2$ kernel uses unit length scales, diagonal regularization $\alpha = 10^{-6}$, normalized outcomes and an improvement offset of 0.01. Design features encode log-scaled learning rate, ordinal weight decay, normalized momentum and the binary residual flag. Both methods have a nominal budget of 700 candidate-rounds. Exhaustive enumeration evaluates all twelve registered configurations, using 1,200 candidate-rounds; this covers the registered menu rather than a full Cartesian hyperparameter grid. The search decisions are replayed using only the development results of queried configurations from the freshly trained, fixed-seed candidate cache. The BioCoLoop rows reuse the original Qwen2.5-7B-Instruct runs, with and without collaborative participation.

Table D.1: **Single-laboratory search controls.** Mean ± sample SD over three training seeds, in percentage units. Search controls use only lab 0; the history-enabled BioCoLoop references use Qwen2.5-7B-Instruct with one or ten laboratories. The ten-laboratory setting adds training and evaluation evidence. Bold marks the highest displayed mean per column.

| Method | Labs | DTI | Proteomics | Cell perturbation | | |
|---|---|---|---|---|---|---|
| | | AUROC↑ | AP↑ | VCC Top-1↑ | Norman Top-1↑ | Tahoe Top-1↑ |
| *LLM-free controls* | | | | | | |
| Fixed model | 1 | 82.83 ±1.22 | 25.05 ±4.56 | 29.94 ±0.00 | **25.56** ±0.00 | 19.40 ±0.00 |
| Random@6 | 1 | 90.10 ±0.71 | 20.13 ±0.06 | 21.10 ±4.41 | 14.10 ±0.04 | 17.91 ±3.95 |
| Bayesian@6 | 1 | 90.10 ±0.71 | 20.01 ±0.24 | 18.61 ±2.78 | 14.12 ±0.04 | 16.92 ±4.80 |
| 12-design menu | 1 | 90.10 ±0.71 | 20.13 ±0.06 | 18.61 ±2.78 | 14.12 ±0.04 | 15.42 ±6.21 |
| *History-enabled BioCoLoop references* | | | | | | |
| BioCoLoop | 1 | 89.89 ±1.02 | 20.13 ±0.07 | 20.22 ±2.22 | 14.15 ±0.00 | 15.42 ±6.21 |
| BioCoLoop | 10 | **93.76** ±1.11 | **36.99** ±0.22 | **31.54** ±0.08 | 22.25 ±0.45 | **22.39** ±2.59 |

The ten-laboratory framework exceeds all three search strategies in mean score on all five endpoints. Relative to random search, the gains are 3.66 AUROC points on DTI, 16.86 AP points on proteomics, and 10.44, 8.15 and 4.48 Top-1 points on VCC, Norman and Tahoe. Including the fixed-model control, BioCoLoop has the highest mean on four of five endpoints; Norman's fixed model retains the best Top-1. These results show the value of the combined collaborative training and research procedure beyond increasing single-laboratory search. Same-access comparisons in Appendix A and Section 4.7 examine the research mechanism separately.

## D.2 Search variability and held-out uncertainty

To assess sensitivity to random proposal choices, we compute the exact expected score of random search over all $\binom{11}{6} = 462$ six-design subsets. The initial design D00 is always included; ties integrate the possible proposal orders using the original development-selection rule. We average the separately scored selected models, not their predictions. Table D.2 reports the expectation for each training seed, summarized across the three seeds. The collaborative framework also exceeds these expected scores on all five endpoints.

For these additional controls, a separate paired bootstrap uses 10,000 resamples of compounds for proteomics and interventions for the cell tasks. Each resample is shared across methods and the

Table D.2: **Integrating random-search variability.** Expected Random@6 score over all $\binom{11}{6} = 462$ proposal subsets, integrating proposal-order ties. This averages selected-model metrics, not their predictions. Scores are percentages; the final row is the mean ± sample SD across three trained seeds after search randomness is integrated out.

| Training seed | DTI AUROC↑ | Proteomics AP↑ | VCC Top-1↑ | Norman Top-1↑ | Tahoe Top-1↑ |
|---|---|---|---|---|---|
| 42 | 88.67 | 20.42 | 22.53 | 15.35 | 14.38 |
| 43 | 90.08 | 20.01 | 19.41 | 15.33 | 12.74 |
| 44 | 89.53 | 20.91 | 20.82 | 15.35 | 19.68 |
| Mean ± SD | 89.43 ±0.71 | 20.44 ±0.45 | 20.92 ±1.56 | 15.34 ±0.01 | 15.60 ±3.63 |

three trained models, and differences are averaged across seeds. These intervals quantify test-group uncertainty conditional on the trained models, complementing the seed standard deviations. The BioCoLoop-minus-random-search 95% intervals are [1.91, 29.97] AP points for proteomics and [−0.48, 22.49], [−9.53, 26.47] and [−3.98, 12.94] Top-1 points for VCC, Norman and Tahoe. The cell intervals span zero despite positive mean differences. These are descriptive intervals without multiplicity adjustment; the original study's uncertainty protocol remains as specified in Appendix A.5.

### D.3 Generalization to unseen DTI entities

The same development-selected checkpoints are evaluated on the original random, unseen-drug and unseen-protein test sets. Exact input-identity checks confirm that none contains a training drug–protein pair; the unseen-drug and unseen-protein sets also have no exact drug-string or protein-sequence overlap with training, respectively. Scores are independently recomputed from saved, row-identified predictions.

Table D.3: **Single-laboratory search across three drug–target test splits.** AUROC and average precision (AP), mean ± sample SD over the same three seeds, expressed as percentages. Each method's development-selected predictor is reused across all three test splits. Lab access, candidate budgets and the historical Qwen BioCoLoop references are as in Table D.1. Bold marks the highest displayed mean per column.

| Method | Labs | Random split | | Unseen drug | | Unseen protein | |
|---|---|---|---|---|---|---|---|
| | | AUROC↑ | AP↑ | AUROC↑ | AP↑ | AUROC↑ | AP↑ |
| Fixed model | 1 | 82.83 ±1.22 | 81.57 ±1.54 | 81.96 ±1.55 | 85.78 ±1.41 | 76.40 ±5.85 | 78.75 ±2.04 |
| Random@6 | 1 | 90.10 ±0.71 | 88.06 ±1.38 | 89.07 ±0.81 | 90.19 ±1.38 | 83.70 ±2.12 | 81.60 ±4.27 |
| Bayesian@6 | 1 | 90.10 ±0.71 | 88.06 ±1.38 | 89.07 ±0.81 | 90.19 ±1.38 | 83.70 ±2.12 | 81.60 ±4.27 |
| 12-design menu | 1 | 90.10 ±0.71 | 88.06 ±1.38 | 89.07 ±0.81 | 90.19 ±1.38 | 83.70 ±2.12 | 81.60 ±4.27 |
| BioCoLoop | 1 | 89.89 ±1.02 | 87.77 ±1.79 | 88.39 ±1.66 | 89.35 ±2.34 | 83.26 ±2.61 | 80.53 ±5.10 |
| BioCoLoop | 10 | **93.76** ±1.11 | **92.73** ±1.81 | **91.65** ±1.64 | **92.58** ±2.16 | **87.63** ±2.50 | **85.07** ±5.34 |

BioCoLoop attains the highest mean AUROC and AP on all three DTI test sets. Compared with the three search controls, its AUROC gains are 3.66 points on the random split, 2.57 on unseen drugs and 3.93 on unseen proteins. The benefit therefore extends beyond the primary random-split endpoint.

The additional study comprises 180 complete candidate fits and 18,000 candidate-rounds. Their recorded fit times sum to 13.02 GPU-hours, excluding feature preparation, held-out scoring and statistical analysis. This is the cost of constructing the shared evaluation cache, rather than the cost of a single search. Aggregate per-seed scores, search definitions and source hashes accompany the tables in `tables/single_lab_search/`.

# Appendix E. Search controls, allocation and validation

## E.1 What the controlled comparison measures

The access scenario in Table 1 allows BioCoLoop to use ten laboratories while the other workflows use one. A same-access search comparison instead gives every method the same ten training laboratories and development panels. The configuration library, initialization, checkpoint selection and proposal/fit budget caps also remain fixed. Random search and Bayesian optimization can use this collaborative fitting service without a language-model researcher. This separates the quality of configuration search from the benefit of accessing additional measurements.

The additional controls reuse the complete twelve-design caches from the six-slot experiment in Appendix B.3: DTI uses training seed 42, and the four lighter endpoints use seed 61. Each candidate has a complete 100-round fit. The corresponding direct and history-guided Qwen/Luna trajectories use the same cache and selection key. Candidate reuse is a retrospective evaluation technique: it preserves the outcome and logical training cost of a query without performing that training again. This experiment is separate from the three-seed access comparison and does not supply additional independent training seeds.

## E.2 Random-search distribution and conventional optimization

Random configuration search provides a direct control for the value of adaptive proposal generation (Bergstra & Bengio, 2012). The fixed model is a one-fit reference; the search controls below receive the same candidate budget as the research loop, rather than that smaller reference budget.

Random search always fits the initial design D00, then selects six distinct configurations from the eleven remaining designs. Enumerating the $\binom{11}{6} = 462$ subsets integrates proposal randomness exactly conditional on the trained candidate cache. Full score/loss ties retain the incumbent; equally ranked newly encountered candidates are averaged over their possible first-encounter orders. A second random control fixes the first proposal to the relevant language-model trajectory and enumerates the remaining $\binom{10}{5} = 252$ subsets. The shared first Qwen/Tahoe slot was invalid: its sensitivity preserves that failed slot and samples five of all eleven remaining designs, giving 462 subsets instead. This distinguishes the shared first proposal from subsequent feedback.

The conventional Bayesian control uses the fixed Gaussian-process expected-improvement recipe in Appendix D, without optimizing that recipe against the held-out data. It observes only the development outcomes of configurations it has queried. Selection probabilities and selected design identifiers are fixed before new held-out scoring. Means of randomly selected model scores are expectations over search decisions, not an ensemble of their predictions. Random-search standard deviations describe configuration-choice variability on one trained cache, not uncertainty over biological samples or independent training seeds.

All search methods have six proposal slots and a 700-round cap including D00. Qwen history on proteomics spends 600 fitted rounds after one invalid slot; both Qwen Tahoe trajectories spend 500 after two. Separate random controls match these realized fit counts. On Tahoe, the four-valid-proposal random expectation is 26.30, compared with 25.37 for both Qwen modes. Fixing a valid shared first proposal gives DTI random expectation 94.43, compared with 94.27 for both history modes. The additional controls therefore retain the conclusion that history is not consistently superior. Source hashes, per-policy selected designs, exact probabilities and cost sensitivities are retained with the aggregate snapshot.

## E.3 Development selection and held-out prediction

We reconstruct development predictions from the already selected checkpoints in the extended VCC and Tahoe searches, rather than refitting or changing the selection rule. The diagnostic covers all recorded prefixes (0, 1, 2, 4, 6, 8, 12, 16, 20 and 24) for both direct and history-guided search. There are 14 distinct selected checkpoints and 140 re-evaluated development panels. Saved held-out ranks independently reproduce all 80 task/backend/mode/prefix scores. All comparisons describe the originally selected models.

Table E.1: **Matched-access configuration search.** All methods use the same ten laboratories, development panels and 12-design library. Searches receive six proposal slots after the common initialization, with up to 700 aggregation rounds. Scores are multiplied by 100. Random search reports its exact expectation and search-policy standard deviation, integrating all 462 six-element subsets; this is not variation across training seeds.

| Method | DTI AUROC↑ | Proteomics AP↑ | VCC Top-1↑ | Norman Top-1↑ | Tahoe Top-1↑ |
|---|---|---|---|---|---|
| Fixed model | 94.27 | **37.27** | **31.50** | 15.48 | 25.37 |
| Random search, no replacement | 94.45 ±0.17 | **37.27** ±0.00 | 27.53 ±6.48 | 22.66 ±3.41 | 26.59 ±0.58 |
| GP–EI Bayesian optimization | **94.60** | **37.27** | **31.50** | 22.30 | 25.37 |
| Qwen: direct search | 94.27 | **37.27** | **31.50** | 22.30 | 25.37 |
| Qwen: history-guided search | 94.27 | **37.27** | **31.50** | 22.30 | 25.37 |
| Luna: direct search | **94.60** | **37.27** | **31.50** | 22.30 | **26.87** |
| Luna: history-guided search | 94.27 | **37.27** | **31.50** | **23.11** | **26.87** |

One frozen training seed per endpoint (42 for DTI; 61 otherwise); agent and GP–EI rows each represent one search trajectory. The fixed reference uses 100 rounds. Failed proposals retain their slots: Qwen history uses 600 fitted rounds on proteomics, and both Qwen arms use 500 on Tahoe; other searches use 700. Matched-realized-fit and shared-first-proposal sensitivities are reported separately. Selection uses development scores only; the comparison reuses checkpoint-identical, fully trained candidates and previously exposed held-out panels. Bold marks the largest displayed mean, not statistical significance.

Table E.2: Development–test divergence in the extended search (training seed 61). Each pair is initialization → 24 proposals. CPU re-evaluation reproduces the original development scores; test scores are independently recomputed from archived query ranks. The equal-laboratory development mean equals the pooled intervention-macro mean because each laboratory contains the same number of development interventions. All scores are macro Top-1 percentages.

| Task | Proposer / search | Development | Held-out |
|---|---|---|---|
| VCC | Qwen / direct and history | 23.65 → 24.83 | 31.50 → 19.61 |
| VCC | Luna / direct | 23.65 → 27.68 | 31.50 → 20.23 |
| VCC | Luna / history | 23.65 → 30.93 | 31.50 → 20.01 |
| Tahoe | Both / direct and history | 30.00 → 35.00 | 25.37 → 13.43 |

In VCC, each of the ten development panels contains two interventions; in Tahoe, each contains six. Their equal-panel macro Top-1 is consequently identical to the pooled intervention-macro Top-1: each development intervention receives weight $1/(10 \times 2)$ or $1/(10 \times 6)$. Different panel sizes in response-query counts do not change this identity, since the primary scorer first averages within an intervention. Thus a discrepancy between these two aggregation rules does not explain the reported decreases.

Development and held-out intervention identities differ. Each intervention also has a fixed five-option roster, sampled within its own split and reused by every method. Repeated checkpoint and configuration selection therefore optimizes a finite development intervention set and its fixed distractor contexts. Better performance on those examples need not transfer to the held-out interventions and rosters. These observations identify a generalization problem; they do not by themselves attribute it uniquely to selection overfitting, biological shift or a particular hyperparameter.

VCC's 20 development interventions generate 302 response queries; its 20 held-out interventions generate 360. Tahoe has one response query per compound (60 development, 67 held-out). Qwen/VCC's development gain of 1.17 points has a descriptive intervention-paired bootstrap interval of $[-20.25, 21.54]$; Tahoe's 5-point gain, equivalent to three additional correct development compounds, has interval $[-11.67, 21.67]$. These post-selection intervals describe the selected models and are not corrected for adaptive search. The primary-first rule can also accept worse loss: Qwen/VCC's development cross-entropy changes from 1.60783 to 1.61182, and Tahoe's from 1.60194 to 1.60877.

**Development-roster sensitivity.** Finite-sample selection criteria can themselves be overfit (Cawley & Talbot, 2010). We test one measurable contributor—dependence on the fixed option roster—without treating that general mechanism as a causal explanation of every observed decrease.

To test dependence on roster construction, we freeze 20 new roster seeds (2026110900–2026110919). Each true development intervention receives four distinct incorrect options sampled within its orig-

Table E.3: Development-roster sensitivity of the already selected initial and 24-proposal final checkpoints. Twenty frozen seeds resample four incorrect options within the same development-intervention pool; each comparison uses paired rosters. Entries are final-minus-initial macro Top-1 differences in percentage points. The standard deviation is across roster draws, not training seeds or new biological splits. No model is trained or reselected.

| Task | Proposer / search | Original roster | New rosters | Positive draws |
|---|---|---|---|---|
| VCC | Qwen / both | $+1.17$ | $-1.34 \pm 7.37$ | $6/20$ |
| VCC | Luna / direct | $+4.03$ | $+2.24 \pm 8.15$ | $12/20$ |
| VCC | Luna / history | $+7.28$ | $+2.89 \pm 7.89$ | $12/20$ |
| Tahoe | Both / both | $+5.00$ | $-1.33 \pm 4.82$ | $7/20$ |

Table E.4: Search with source-defined laboratories. Protein: PTPC, Lin, Ruprecht and decryptE; cell: VCC, Replogle, Nadig and Jiang. All methods share four training sources and the target development panels. Searches use twelve designs, six proposal slots and at most 700 rounds; the fixed reference uses 100 rounds. Mean $\pm$ sample SD over training/search seeds 61–63; scores $\times 100$. Random reports the exact policy expectation per seed, then its between-seed SD. Luna uses the same checkpoint cache but new proposal calls.

| Method | Proteomic AP $\uparrow$ | Cell macro Top-1 $\uparrow$ |
|---|---|---|
| Fixed design | $38.72 \pm 1.94$ | $28.82 \pm 0.00$ |
| Random, exact expectation | $36.37 \pm 1.27$ | $24.83 \pm 4.23$ |
| Qwen direct | $37.83 \pm 1.28$ | $22.36 \pm 7.20$ |
| Qwen history | $34.31 \pm 4.96$ | $19.41 \pm 2.11$ |
| Luna direct | $35.35 \pm 3.62$ | $22.88 \pm 6.75$ |
| Luna history | $34.94 \pm 3.52$ | $22.88 \pm 6.75$ |
| Luna enriched history | $35.39 \pm 3.67$ | $22.88 \pm 6.75$ |

inal development pool, and the five option positions are shuffled; all response queries for that intervention share the roster. Initial and already selected final checkpoints receive identical rosters. This study neither trains nor reselects a model and does not read test responses or scores.

Qwen/VCC's original advantage is positive in only 6 of the 20 new rosters; Tahoe's in 7. Luna/VCC retains smaller positive mean advantages, with 12 positive draws for each mode. The standard deviations quantify roster randomness on the same interventions, not independent training or biological replication. The result establishes sensitivity to distractors and option order; a new intervention-disjoint validation experiment would be required to separate it from biological distribution shift and repeated-selection effects.

### E.4 What source-defined laboratories establish

The source-defined studies in Appendix B.2 are distinct from splitting one corpus into ten clients. The proteomics experiment uses PTPC, Lin, Ruprecht and decryptE as four training laboratories, with shared encoders and source-specific prediction heads. Its fixed-design target AP increases from 34.41 to 37.53 with all three auxiliary studies. The matched cell experiment instead compares four VCC groups with four source-defined laboratories (VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562), each with 15 training interventions. The latter scores 28.82 macro Top-1 versus 30.70 for the same-source groups. Cell counts and measured genes remain source-specific.

These comparisons test actual differences between released datasets, not independent administrative deployment. They establish beneficial and negative source transfer under a common target scorer, not universal improvement from heterogeneity. To assess history under these sources, we additionally report the completed twelve-design caches for both tasks and three training seeds, with direct and history-guided search at matched source access (Table E.4). The fixed-design reference in this three-seed study is specific to its cache and must not be substituted for the separate seed-61 transfer contrast above.

The source-defined comparisons also do not show a general history advantage: Qwen history scores below direct on both tasks. A subsequent Luna study adds explicit expected-versus-observed feedback and configuration-change summaries to its history. Protein AP increases from 34.94 to 35.39 relative to the original history, but direct already reaches 35.35; all three cell trajectories select the

same checkpoint within each seed. This enriched-context intervention changes multiple prompt fields and token costs, so it is not a single-field ablation. All 72 historical candidate predictions were reverified; new proposal calls are distinguished from new predictor training.

In these source-defined adapters, target evaluation first averages the ten original target panels inside one target worker. A second aggregation preserves this primary mean but records one returning worker, so its recorded `worst_client` equals the already aggregated primary score. It is not a recovered worst-panel diagnostic. This does not change the target selection score; it limits how much cross-panel disagreement is available to the proposal model. The main ten-client protocol and this nested source-study representation are therefore documented separately.

### E.5 Comparing training-promotion rules

The allocation result in Section 4.4 compares screening and promotion with uniform 80-round training. To examine the role of the promotion rule itself, we additionally compare the original score-plus-trajectory policy with three alternatives: promoting the six best screening scores globally, promoting the two best scores within each learning-rate group, and promoting six uniformly sampled designs. Screening comparisons use the best development primary score followed by lower development loss; complete screening ties retain the original slate order. Full-training selection retains the first promoted candidate on a complete score/loss tie. The trajectory policy retains the unchanged rule in Section 3.4.

All policies use the same ten-design slate, ten training laboratories, target development panels and initialization rules. Each promotion policy pays for ten 20-round screens and six fresh 100-round fits; the uniform policy pays for ten 80-round fits. Both use 800 aggregation rounds and 160 aggregate development evaluations. These are matched training/evaluation counts; runtime can differ with the selected designs. We newly train every design at each of the three durations for four tasks and seeds 53–55, yielding 360 complete fits and 24,000 actual aggregation rounds. This complete cache supports retrospective policy evaluation under a common runtime; the per-policy budget is distinct from the total cost of constructing it. Each duration is fitted from initialization, rather than obtained by truncating a longer run.

Table E.5: Fresh-runtime allocation controls. Scores $\times 100$ are mean $\pm$ sample SD over training seeds 53–55. All policies use the same ten laboratories and ten designs, with 800 aggregation rounds and 160 aggregate development evaluations. Promotion policies screen ten designs for 20 rounds and restart six for 100; uniform allocation fits all ten for 80. Random promotion integrates all 210 six-design subsets per seed, and its displayed SD is across seed-conditional expectations. All 360 fits are newly trained; the original Table 2 is retained separately. Bold marks the best unrounded mean, including ties.

| Allocation policy | Proteomics AP ↑ | VCC Top-1 ↑ | Norman Top-1 ↑ | Tahoe Top-1 ↑ |
|---|---|---|---|---|
| Uniform (80 rounds) | **37.47** $\pm$ 0.55 | 20.01 $\pm$ 0.00 | 11.48 $\pm$ 0.52 | 20.90 $\pm$ 2.99 |
| Random promotion | 37.37 $\pm$ 0.48 | **22.49** $\pm$ 1.70 | **22.63** $\pm$ 0.64 | 21.95 $\pm$ 0.04 |
| Score only (global) | **37.47** $\pm$ 0.55 | 20.01 $\pm$ 0.00 | 21.70 $\pm$ 7.19 | 20.90 $\pm$ 0.00 |
| Score only (LR groups) | **37.47** $\pm$ 0.55 | 20.01 $\pm$ 0.00 | 21.70 $\pm$ 7.19 | 20.90 $\pm$ 0.00 |
| Score + trajectory | **37.47** $\pm$ 0.55 | 20.01 $\pm$ 0.00 | 22.32 $\pm$ 0.43 | **22.89** $\pm$ 1.72 |

All development-selected models are sealed before new held-out scoring. Random promotion integrates all $\binom{10}{6} = 210$ subsets, using the original slate order to resolve final-selection ties. Its outcome is the expected score of a selected single predictor. The table's random-row SD is across the three seed-conditional expectations; within-seed search-policy variation is recorded separately. These are the original biological partitions, not new external validation populations. Table 2 remains the original allocation study, while Table E.5 reports this fresh-runtime comparison.

The fresh run reproduces all four original uniform and trajectory-policy means. Against uniform allocation, the trajectory policy again yields four wins, eight ties and no losses across the twelve task–seed pairs. Against either score-only policy, it yields four wins, seven ties and one loss. Against the exact random-promotion expectation, it yields six wins and six losses. These counts summarize paired directions, not a pooled effect across different task metrics. Global and learning-rate-stratified score-only promotion give the same task means here. The trajectory policy's Norman mean is 0.62 points

higher than their 21.70, but below random promotion's 22.63; on Tahoe it exceeds both controls. On VCC, random promotion instead scores 22.49 versus 20.01. Much of the Norman gain over uniform allocation is thus also achieved by simple promotion schedules that fully train fewer candidates; the particular trajectory-based rule adds a task-dependent effect.

### E.6 History representation on a larger gene-disjoint study

The preceding controls motivate a fresh experiment that separates feedback representation from access and fitting budget. It uses two source studies, Replogle and Nadig, with four study–cell contexts: K562, RPE1, HepG2 and Jurkat. Each context acts as a training and evaluation laboratory. These are four biological contexts, not four independently deployed institutions. After vocabulary and measurement filtering, the training, development and held-out sets contain 3,806, 1,247 and 1,282 conditions, respectively, covering 1,208, 414 and 406 distinct perturbation genes. Gene names and model-vocabulary identifiers are disjoint across the three partitions.

**Predictor and measurement.** We freeze the corrected scDEBART encoder and select 512 response genes by training-response variance, requiring measurement in all four contexts. The predictor receives control expression and the candidate perturbation identifier. It predicts the difference between perturbed and control mean log-normalized expression. Evaluation identifies the true perturbation among five options within a context. Each gene–context is evaluated on five fixed rosters; every option uses the same mask, excluding the five candidate perturbation genes wherever they occur in the response panel. The primary score first averages rosters within a gene, then genes within a context, and finally the four contexts equally. Repeated rosters do not create independent biological replicates. This new response panel and source-defined benchmark are separate from the original VCC, Norman and Tahoe endpoints.

**Common candidate and fitting protocol.** The registered library has 16 designs: four learning rates ($10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}$), two prediction heads (fine-tuned pretrained head or frozen head plus a zero-output-initialized residual adapter), and two objectives (response MSE or MSE plus 0.1 times within-batch contrastive cross-entropy). Weight decay is 0.001 and coordinator momentum is zero. All 48 design–seed fits start afresh and complete 20 full-client rounds with batch size 32. Development checkpoints at rounds 2, 5, 10 and 20 are ranked by Top-1, then lower cosine-ranking cross-entropy and earlier round. The pretrained round-0 predictor is diagnostic only. The fixed D00 reference is the fully trained head with learning rate $10^{-4}$ and MSE, not the untrained initialization.

**Matched feedback conditions.** Qwen2.5-7B-Instruct generates six candidate requests after D00, with the first request shared across three modes. Direct receives tried identifiers without scores; raw history receives the observed development scores, trajectories and context diagnostics; structured history receives the same facts organized around reference configurations, exact configuration changes and expected-versus-observed effects. Neither history mode observes an unqueried candidate. Proposals contain `design_id`, `reference_id`, `changed_fields`, the hypothesis, experiment and expected effect, and a numeric prediction. The executor checks identifiers and configuration changes against the registered library; those fields, rather than the rationale text, determine execution. This makes the proposal's stated comparison explicit without assigning credit merely for predicting an outcome correctly.

The final study reruns all three modes after an earlier output parser rejected otherwise valid JSON wrapped in code fences or containing literal backslashes. The same limited formatting normalization is applied to every mode; malformed schemas and incorrect configuration changes still fail, with at most two generation attempts per slot. The earlier run remains an engineering diagnostic, not an additional scientific replicate. Direct and structured history each complete all 18 slots across seeds 61–63; raw history completes 15. Conventional random search and GP–EI use the same candidate cache and development-selection key. Random search exactly integrates the $\binom{15}{6} = 5,005$ subsets, with shared-first and realized-fit-count sensitivities retained in the result record.

**Outcomes and resource accounting.** Structured history selects a different final design from direct in two of three seeds. Its mean held-out Top-1 is 22.888%, versus 22.625% for direct and 22.668% for random search; GP–EI reaches 23.152%. Structured-minus-direct is $+0.263$ points (two gains and one tie), with paired gene-cluster 95% interval $[-1.022, 1.568]$. Raw history and direct select

Table E.6: Fresh history-feedback comparison on the gene-disjoint, four-context cell benchmark (three paired training/controller seeds). This benchmark is separate from the original VCC, Norman and Tahoe tasks. Top-1 scores are percentages; ± reports the sample standard deviation across seeds. Search methods share the same 16 designs, four clients, development panels and six proposal slots; each valid candidate is fitted for 20 rounds. Fits include the initial-design fit. Random search integrates all size-six subsets of the 15 non-initial designs and selects using development evidence, not test scores. Raw history has three invalid slots across 18; direct and structured history complete all 18. Bold marks the best mean development or test score.

| Search method | Development | Held-out | Fits |
|---|---|---|---|
| Fully trained fixed D00 | $22.40 \pm 0.07$ | $21.43 \pm 0.10$ | 1 |
| Direct (no feedback) | $22.68 \pm 0.25$ | $22.62 \pm 0.21$ | 7 |
| Raw history | $22.68 \pm 0.25$ | $22.62 \pm 0.21$ | $6 \pm 1$ |
| Structured history | $23.38 \pm 0.48$ | $22.89 \pm 0.56$ | 7 |
| Random without replacement | $23.06 \pm 0.24$ | $22.67 \pm 0.21$ | 7 |
| GP–EI | $\mathbf{23.60 \pm 0.32}$ | $\mathbf{23.15 \pm 0.35}$ | 7 |
| *Paired held-out contrasts (percentage points)* | | | |
| Contrast | Mean difference | 95% CI | W/T/L |
| Structured − Direct | $+0.263$ | $[-1.022, 1.568]$ | 2/1/0 |
| Structured − Random | $+0.220$ | $[-0.290, 0.738]$ | 2/0/1 |
| Structured − GP–EI | $-0.264$ | $[-0.930, 0.413]$ | 0/2/1 |

W/T/L counts paired seeds. Confidence intervals use 10,000 paired perturbation-gene bootstrap resamples and describe conditional test-gene uncertainty for the fixed selected models, rather than uncertainty over additional training or controller runs.

Table E.7: Resource accounting for the fresh history comparison. Training columns are per-seed means; the last two columns total all three seeds. Fits are evaluated from a shared cache, with each policy charged for every candidate it requests. Raw-history invalid slots consume proposal budget without a candidate fit. Identical rounds, steps and examples do not imply identical FLOPs or language-model costs. Calls include permitted retries and logically charge the shared first proposal to each language-model arm.

| Method | Rounds | Steps | Examples | Invalid slots | LLM calls |
|---|---|---|---|---|---|
| Fully trained fixed D00 | 20 | 2,420 | 76,120 | 0 | 0 |
| Direct (no feedback) | 140 | 16,940 | 532,840 | 0 | 18 |
| Raw history | 120 | 14,520 | 456,720 | 3 | 24 |
| Structured history | 140 | 16,940 | 532,840 | 0 | 21 |
| Random without replacement | 140 | 16,940 | 532,840 | 0 | 0 |
| GP–EI | 140 | 16,940 | 532,840 | 0 | 0 |

the same final predictor in every seed. The confidence intervals use 10,000 global-gene resamples shared across contexts, methods and seeds, conditional on these fitted and selected predictors. They measure held-out gene uncertainty, not variation over further controller runs. The results show that the feedback representation can change search decisions, but do not establish stable superiority over direct or conventional search.

Every valid fit uses 2,420 optimizer steps and 76,120 training-example exposures. Direct, structured, random and GP–EI therefore each use 16,940 steps and 532,840 exposures per search. These matched counts do not imply identical FLOPs: the head choices have different trainable parameter counts. Logical language-model calls total 18 for direct, 24 for raw and 21 for structured, including retries; generating the shared first proposal only once gives 57 physical calls overall. Training/development feature extraction and all 48 predictor fits cost approximately 0.236 GPU hours, excluding language-model inference and held-out evaluation. This construction cost is distinct from each policy's logical budget. Held-out scoring follows the frozen development selections and independently verifies all 48 models.

**Development and held-out trajectories.** Figure 6 reports every prefix from zero to six requests for every policy and seed. All final search outcomes improve on their corresponding fitted D00 reference, so this study does not reproduce the earlier large terminal decreases. It does retain a local reversal: in seed 62, raw and structured history select D05 at slot 2, raising development Top-1 by 0.057 points while lowering held-out Top-1 by 0.257; later selections recover. The K562 context also has a positive mean development change but a negative mean held-out change under structured history. Its largest overall development improvement occurs at the final slot, and its prefix-average development score remains below random search and GP–EI. The new data, response panel, design library and feedback format change together relative to the old benchmark, so the improved terminal behavior is not a controlled attribution to any one change. The diagnostic scores the originally selected prefix models; it does not choose a stopping point using held-out outcomes.

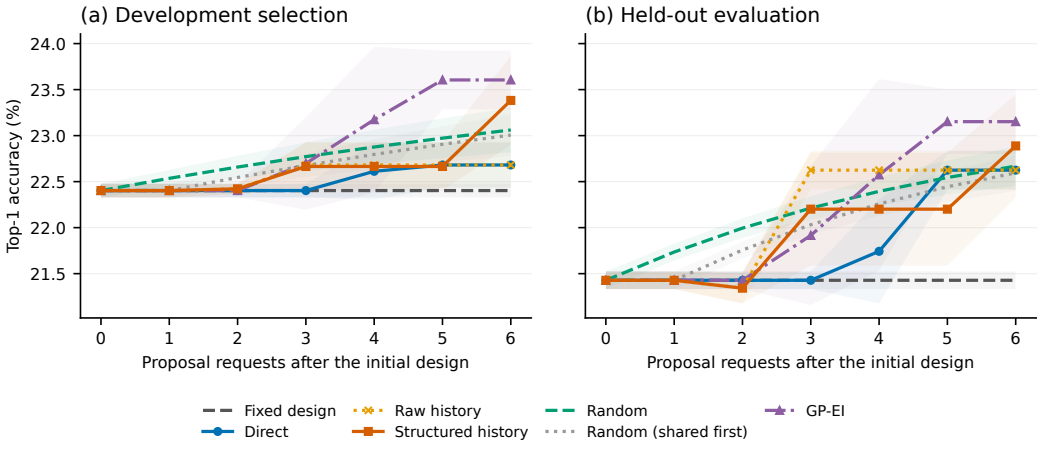



Figure 6: Development selection and held-out prediction across the complete six-request study. Lines are means across three paired seeds and shading is sample standard deviation. Random curves integrate development-selected single-model outcomes at each budget, rather than depicting one random trajectory. The fixed reference has received its full 20-round fit. Held-out trajectories evaluate the development-selected model at each prefix, not the best model according to test scores.

### E.7 Sensitivity to the single participating laboratory

To examine whether laboratory 0 represents independent laboratories generally, we additionally train the fixed D00 design separately on every laboratory, 0–9, for proteomics, VCC, Norman and Tahoe. Each fit uses seed 42, 100 rounds, only that laboratory's training data and only its development panel for checkpoint selection. All 40 models, including laboratory 0, are newly fitted under the same CPU runtime and sealed before evaluation on the common task-specific test set. Saved predictions independently reproduce all 40 primary scores. The standard deviation here describes variation across laboratory identities at one seed, not training-seed variation or a collaborative model.

Table E.8: Fixed-design sensitivity to the identity of the single participating laboratory. Each of ten laboratories independently fits D00 for 100 rounds and selects a checkpoint on its own development panel (seed 42). All 40 fits are fresh CPU runs under the same training equations; the SD and range describe laboratory identities, not training seeds. This diagnostic does not rerun research controllers.

| Endpoint | Lab 0 | Single-lab mean $\pm$ SD | Range |
|---|---|---|---|
| Proteomics AP | 24.01 | $28.69 \pm 6.44$ | 14.69–37.34 |
| VCC Top-1 | 29.42 | $32.50 \pm 3.49$ | 26.96–39.76 |
| Norman Top-1 | 25.56 | $17.67 \pm 5.84$ | 4.37–25.56 |
| Tahoe Top-1 | 19.40 | $19.70 \pm 4.66$ | 13.43–26.87 |

Laboratory 0 is below the independent-laboratory mean for proteomics and VCC, close to it for Tahoe, and the highest-scoring laboratory for Norman. Thus its representativeness is task-dependent. These results supplement, rather than replace, the original three-seed GPU workflow scores. They evaluate fixed-model identity sensitivity, not a rotation of every search controller across laboratories. DTI is not included: its present preprocessing uses a training dictionary constructed from laboratory 0, so an isolated-laboratory rotation would also require a laboratory-specific dictionary and a separately registered initialization protocol. The same-ten-laboratory search controls in E.1–E.2 remain the direct comparison of search methods at fixed access.